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# Understanding Customer Behaviours and Perception of Emerging Speaker Connectivity Technologies

Enhancing Speaker User Experience by Exploring Knowledge and Awareness of Connectivity Solutions Through a User Study

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## Abstract

This thesis explores how emerging connectivity technologies can be integrated into speaker systems to improve the overall user experience. First, a pre-study was conducted to establish an understanding of the market which Marshall operates within and its brand identity. This was followed by a user study, consisting of a survey, observations, interviews and a workshop. These were conducted to identify user behaviours, needs and pain points. The findings revealed that users prioritise simplicity, familiarity and intuitive interaction, while experiencing challenges related to connectivity and system understanding.

Based on these insights, the design concept Bexley was developed, consisting of a movable speaker and a centralised docking station. Further the insights indicated that the user interface of the application could be enhanced to increase understanding and value for the user. The concept aims to reduce user friction and create a cohesive ecosystem that integrates both physical and digital interactions. Particular focus was placed on maintaining Marshall's distinctive design identity while introducing new features.

The results demonstrate the importance of aligning technological developments with user expectations, highlighting that successful solutions must not only be technically advanced but also easily understood.



#### Keywords

Portable Audio Technology, Audio Device Connectivity, User Study, Product Development, Interactive Feedback, User Interface

## Sammanfattning

Denna studie undersöker hur nya uppkopplingsteknologier kan implementeras i högtalarsystem för att förbättra användarupplevelsen. Först genomfördes en förstudie för att skapa förståelse för marknaden där Marshall verkar samt förstå företagets varumärkesidentitet. Därefter genomfördes en användarstudie, bestående av en enkät, observationer, intervjuer och en workshop. Detta i syfte att identifiera användarbeteenden, behov och utmaningar. Resultaten visade att användarna värdesätter enkelhet, igenkänning och intuitiv interaktion, samtidigt som de upplever utmaningar kopplade till uppkoppling och systemförståelse.

Baserat på dessa insikter utvecklades konceptet Bexley, bestående av en flyttbar högtalare och en central dockningsstation. Dessutom visade resultatet att en utveckling av användargränssnittet kan öka förståelsen och värdet för användaren. Konceptet strävar till att minska användarfriktion och skapa ett sammanhängande system som integrerar både fysisk och digital interaktion. Ett särskilt fokus låg även på att bevara Marshalls identitet samtidigt som dessa nya funktioner introducerades.

Resultatet belyser vikten av att anpassa teknologiska utvecklingar efter användarnas förväntningar och behov, samt visar att framgångsrika lösningar inte enbart behöver vara tekniskt avancerade, utan också lätta att förstå.



#### Nyckelord

Portabel ljudteknik, Ljudenhetsanslutning, Användarstudie, Interaktiv återkoppling, Användargränssnitt

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## List of Acronyms and Abbreviations

| | |
|---|---|
| Bluetooth LE | Bluetooth Low Energy |
| NFC | Near Field Communication |
| PSU | Power Supply Unit |
| PU | Polyuretan |
| UI | User Interface |
| UX | User Experience |

# 1. Introduction

This chapter presents an overview of the degree project by providing background and problem description. It states the underlying purpose of which the degree project stems as well as relevant delimitations that clarifies the scope of the study. The project was conducted in collaboration with Marshall.

The project explored users' experience and perceptions of speaker interactions, with the aim to identify how these can be designed to support user understanding and interaction. Particular focus was placed on how emerging technologies could simplify interactions, without introducing new friction, while maintaining Marshalls brand identity and a tactile user experience.

Based upon a broad user study consisting of a survey, two observations, interviews and a workshop. These insights laid the foundation for the concept development process. The proposed solution resulted in a portable speaker, a multifunctional docking station, and an accompanying application.

## 1.1. Background

Marshall was founded by Jim and Terry Marshall, father and son, in London in 1962, by initially producing amplifiers. Marshall wanted their sound to be defined by rock 'n' roll. Over time, Marshall have expanded their portfolio, and the general public know them for their speakers and headphones. While the company originally focused on providing high quality sound equipment for musicians, they have expanded their target group to consumer audience, but still focusing on delivering high quality sound. The company's slogan "We are home of loud" reflects this identity (About Marshall, n.d).

Speakers can broadly be divided into two main categories: portable and stationary. Portable speakers could be further divided into those for on-the-go or those intended as movable in and around the home. The main categories, in turn, typically differ in the technologies they rely on. Portable speakers commonly use Bluetooth connectivity while stationary speakers often rely on Wi-Fi connectivity. Bluetooth does not require an internet connection making it well suited for portable speakers. Wi-Fi speakers require an internet connection, which limits their use to locations where Wi-Fi is available and therefore makes them a common choice for stationary set-ups, such as office environments or private homes.

Marshall's portfolio consists of mainly Bluetooth speakers, operated mainly by tactile and manual buttons. Marshall stays relevant by continuously integrating new technology into their products, while still keeping their retro aesthetic, features and brand image intact. However, the company does not aim to position themselves as a frontrunner in technological innovations. In contrast to some competitors which strongly emphasise the integration of new technologies and application-based features as an unique selling point.

## 1.2. Problem Description

The goal is to examine how Marshall could integrate technologies for connectivity seamlessly into the speaker experience, without creating any new frictions. Is there a way to integrate new

technology in a way that preserves the brand identity while bringing added value to both current and potential customers?

### 1.3. Purpose

The purpose of this study was to examine and understand user behaviours and perceptions, through a user study, surrounding speakers. Based on insights gathered from the user study, a solution was developed to improve the user experience. However, the development of the was not predetermined by the company, rather it emerged from the results of the user study.

The user study will be guided by three user cases:

I. Situations where multiple speakers are intended to sync and play simultaneously

II. Situations where multiple users wish to play music together on the same speaker(s)

III. Situations where a speaker is used across different locations, both in and around the home and potentially outside the home

### 1.4. Delimitations

To delimit the scope of the project, the study focuses solely on portable speakers rather than the full spectrum of speakers. Furthermore, the project does not aim to develop new technologies, rather it builds upon existing solutions and techniques. The study is further delimited to three predefined user cases, thereby excluding other potential usage scenarios.

## 2. Frame of reference

In the following chapter, background research on different technologies for synchronised playback are explained. Additionally, research on interaction design was conducted to explore how products should be developed to enhance users comprehension and be straightforward to use. Lastly, theories and perspectives on how humans interpret and adopt new technologies are addressed.

### 2.1. Synchronised Playback Technologies

Modern speaker systems primarily rely on two dominant wireless connectivity methods: Bluetooth and Wi-Fi. Although both enable wireless audio transmission, they have an unique set of characteristics such as differences in range, sound quality and compatibility with other devices (Sonos, n.d.).

Wi-Fi speakers range is equivalent to the reach of the wireless network but have the ability to transfer large numbers of data resulting in lossless audio. Bluetooth connections on the other hand are usually limited to approximately 10-20 meters between the audio-streaming device and the speaker. This direct wireless, often one-to-one, connection is a short-range radio signal that is ideal for portable use (Philips, n.d.). Wi-Fi speakers require more power consumption when transmitting data compared to Bluetooth connection, making portable speakers frequently rely

on solely Bluetooth and stationary often include both connectivity methods. When streaming music via Bluetooth the data is compressed when transferring the audio to the speaker, this can affect the audio quality and user experience. Advancements in Bluetooth technology have enabled the transfer of lossless audio, thus allowing Bluetooth devices to deliver a greater music experience. However, compared to Wi-Fi which benefit from broader bandwidth capabilities, Bluetooth still has certain limitations in high-quality audio streaming (Sonos, n.d.). At the same time, Wi-Fi based audio systems remain dependent on the stability and quality of the local network connection. which may vary between environments and affect user experience.

Multiroom speaker systems consist of multiple speakers synchronised to provide cohesive audio experience throughout multiple rooms. Traditionally, Bluetooth technology supports pairing of one-to-one device connections, as seen in Figure 1, although multi connection solutions have arisen recently. While it is possible to synchronise multiple Bluetooth speakers, such configurations are generally constrained by range limitations and scalability. Wi-Fi allows for one-to-many connectivity, thus making multiroom speakers systems dominated by Wi-Fi dependent solutions. Figure 1 visualises a simplified explanation of Bluetooth and Wi-Fi. The introduction of Bluetooth Auracast expands the potential for employing Bluetooth speakers in multiroom systems (Burger, 2025).

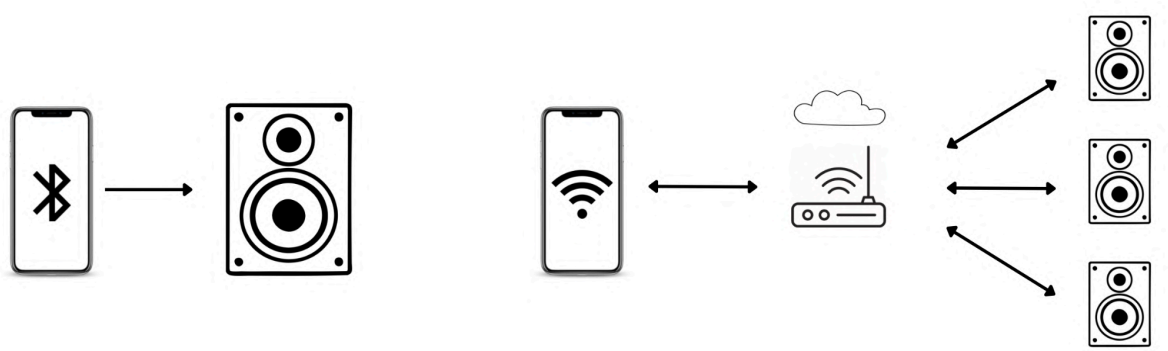


*Figure 1: Visualisation of Bluetooth and Wi-Fi*

Auracast is based on Bluetooth Low Energy (BT LE) Audio and consists of two primary components: an Auracast transmitter and an Auracast receiver. A device equipped with a transmitter, can broadcast audio that can be received by any compatible device containing an Auracast receiver (Auracast, n.d.). At present many smartphones lack an Auracast interface. Consequently, communication between the smartphone and the Auracats transmitter must be facilitated via a unit through alternative technologies, such as Wi-Fi or Bluetooth. The receiver listens to the transmitted audio stream without establishing a traditional one-to-one Bluetooth pairing. As a result, an unlimited number of receivers can join the same audio broadcast simultaneously. Figure 2 visualises a simplified version of how Auracast works. Under favorable conditions, Auracast can reach distances of up to approximately 100 meters, in a spherical field, in open environments (Auracast, n.d.)

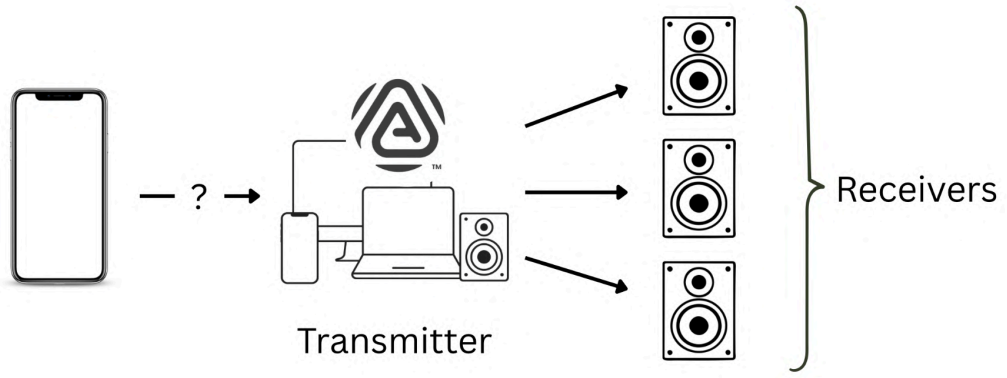



*Figure 2: Visualisation of Auracast*

Moreover, in environments where multiple Auracast broadcasts are available simultaneously, an additional component referred to as an Auracast assistant, may be used. The Auracast assistant provides a user interface that enables navigation between available broadcasts, similarly to choosing a Wi-Fi network from a list. Importantly, the assistant does not transmit or receive the audio itself, but instead facilitates the initial discovery, selection and configuration of the broadcast (SIG Staff, 2024).

A broadcast may be either open or encrypted. Open broadcasts are accessible to any compatible receiver within range, making them suitable for public environments such as gyms, airports or restaurants. In contrast, private use cases, such as listening to television in a domestic setting, may require encryption to prevent unauthorised access. Encrypted broadcasts can be accessed with a key, called Broadcast_Code, which functions comparable to a Wi-Fi password. Exactly how the Broadcast_Code is entered is left to implementation (SIG Staff, 2024).

## 2.2. Interaction design

Literature on interaction design was reviewed to develop an understanding of how products can be designed based on users behaviour and how meaningful interaction can be created.

Affordance refers to the relationship between an object and its user, describing the action that a user can perform with the object (Norman, 2013). For example, a speaker affords the output of sound. To communicate an object's capabilities, what it affords, the object must be provided with indicators that make these affordances perceivable to the user. These indicators are referred to as signifiers, and may take various forms, such as words or symbols. In some cases, an affordance may be sufficiently apparent to function as its own signifier. While both affordance and signifiers are essential in products, signifiers are particularly important from a design perspective, as they communicate how the product is intended to be used and guide user interaction.

As technology advances, user interactions increasingly occur through digital systems, where affordance and signifiers often are communicated through user experience (UX) and user interface (UI). UX refers to the overall experience of a user's interaction with a product, while UI

refers to the visual elements a user encounters interacting with during the user experience (Berghs, n.d.). Both UX and UI influence how users perceive, navigate and interact with products (Berghs, n.d.).

The importance of UX and UI has grown significantly more important, as they are viewed as tools that enable interaction between the consumer and the brand (Dhingra and Shetty, 2024). Intuitive UX and UI design facilitates user navigation and interaction, thereby enhancing the overall perception of a brand and increasing the likelihood of continued engagement (Dhingra and Shetty, 2024). A well-designed interface should support users in understanding which actions to perform, as well as when and how to perform them (Sharma and Tiwari, 2021).

Designers communicate intended actions through the careful use of typography, colour and graphical elements as well as their arrangement. Improved user experience can be achieved through research, the analysis of existing user experiences, and iterative refinement based on user feedback (Sharma and Kumar Tiwari, 2021).

Defining what constitutes an intuitive and effective interface is challenging, as design trends continuously evolve (Gatsou and Farrington, 2021). However, a clear and minimalistic design has been proven to support effective user interaction with digital systems (Gatsou and Farrington, 2021).

One example of a design trend is skeuomorphism, which was prominent in the mid 2000s. Skeuomorphism is when real life objects are imitated in a digital interface to create familiarity. It was particularly popular when touch screens were first introduced, as it helped users understand how to interact with digital elements. Today, skeuomorphism elements are used more sparingly, but still used in example toggle switches and sliders, as fully skeuomorphic interfaces are often considered visually cluttered. Instead, skeuomorphism is primarily used as a stylistic design choice to convey brand identity (Gatsou and Farrington, 2021).

## 2.3. Humans and New Technologies

The adoption of new technologies is strongly influenced by user's perception rather than functionality alone. As some researchers state, negative impressions tend to outweigh positive ones, indicating that uncertainty and lack of understanding can significantly hinder acceptance (Frank et al., 2022). How to present and market emerging or implemented technologies is therefore of much relevance, where unfamiliar interaction patterns may increase perceived risk.

Technology adoption models further identify perceived usefulness and ease of use as two key determinants of acceptance (MIT, 2020; Silhavy and Silhavy, 2024). These factors are especially important for users with limited technical experience as they influence motivation and ability to engage with new systems. In addition to physical interaction, digital interfaces such as mobile applications play a central role in technology adoption. Applications are often used to access advanced features, manage connectivity, or configure devices which can introduce additional complexity if not designed correctly. Poorly designed interfaces may therefore increase cognitive

load and negatively affect perceived ease of use, while well-designed systems can support understanding and reduce barriers of adoption.

At the same time, increased functionality often leads to greater interface complexity. As noted by Carbon (2019), many usability challenges persist because products include too many functions and controls. This highlights an ongoing challenge in product design, balancing functionality richness with usability.

# 3. Methodology

This chapter presents the methods applied in this study. The research process is guided by the Design Thinking mindset, specifically structured around the Double Diamond framework. The tools used within the Double Diamond framework are acquired from Openlab's toolkit (Openlab Stockholm, n.d) along with tools from additional valuable sources. AI tools have been used to improve the language of the report with the prompt "Provide examples of how this text can be refined to improve readability and suit an academic master thesis". However, all content has been produced by the authors.

## 3.1. Design Thinking and Double diamond

The design thinking mindset was applied in this project. Design thinking helped determine which activities to carry out, when to perform them, and how long to spend at different stages before moving on to the next (Liedtka, 2018). Most importantly, design thinking supported an iterative process, allowing for revisiting and reviewing previous steps.

The Double Diamond, which originates from design thinking, is a framework divided into four phases that guide the progression of a design project. The four phases are; Discover, Define, Develop and Deliver. The first phase, Discover, consists of diverging research in order to understand the problem. The research conducted in this phase, aims to understand the underlying problems by gathering insights from people affected by the issues (Design council, n.d.). The second phase, Define, focuses on converging and synthesising the collected data from the earlier stages in order to articulate a well-grounded problem definition. The Develop phase is the third stage, where divergent thinking is applied to explore multiple solutions based on the clearly defined problem. Lastly Deliver, is about testing the different ideas and narrowing down to the ones that may work.

## 3.2. Competitive Analysis

For a company to successfully compete within an industry, it is essential to understand the environment the company operates within (Deshpandé and Gatignon, 1994; Kim et. al, 2025). Such information can be gathered from a variety of sources, including published materials and documentation, observations, as well as insights from current or former employees working with competitors (Deshpandé and Gatignon, 1994).

In this study, information was collected by analysing online reviews, both general reviews of the companies and product-specific reviews important to this study, such as portable speakers. Following this, observations were conducted at several resellers, where each competitor's store display was examined, alongside interviews with store employees. Based on the information gathered, frequently highlighted themes and recurring issues were identified. Lastly, brand mapping was carried out, based on the customer's perceived image of each company. This approach portrayed not what the companies claim to stand for, but how their customers actually view them.

## 3.3. Quantitative Survey

The quantitative research was conducted by sending out a survey that contained questions regarding speaker usage amongst people in their everyday life. The survey consisted of 12 questions and a combination of closed-ended and open-ended questions were used to both ease the gathering of data as well as allowing respondents to form answers without predetermined formats. This affirms the respondent and prevents negative feelings of the survey (Eichhorn, 2024). The closed-ended questions were later coded linking an answer directly to one discrete answer. An example of a closed-ended question was *"With whom do you live?"* where the answers were *"Alone"* or *"Not Alone"*. An example of an open-ended question was following a question on what type of speaker the user has at home at the moment "*Why did you choose this type of speaker?*" and the respondents could write a short answer on their own. There were also questions with a combination of closed- and open-ended where the respondents could choose a predetermined answer but also choose to write on their own if needed. Furthermore the survey included multiple-choice questions like *"What is most important to you in a speaker system?"* where the respondents could choose multiple answers such as *"Soundquality"*, *"Price"*, *"Brand"* etc. This aimed to capture that what users value could be multi-dimensional (Eichhorn, 2024). Lastly a scale-question was asked where the respondent got to answer "*How technologically confident would you say you are?"* ranking themselves from 1- *Not confident at all* to 5 - *Very confident*.

Before sending the survey to a mass public, it was sent to be tested if the formulations and questions were good and understandable. After the feedback, minor alterations were done and the survey was posted.

## 3.4. Observations

Observations were performed to capture behavioural patterns by investigating what users actually do, rather than relying solely on what they say. The method ensured a deeper understanding of the customer and enabled identification of unforeseen behaviours, tacit interactions (Openlab Stockholm, n.d.). Two observations were conducted to examine the user cases.

During the installation, written notes and photo documentation were collected, to capture interaction sequences, repeated behavioural patterns and unexpected actions. An observation sheet was prepared in advance, containing key behaviours and indications of friction. This sheet acted as a guide during the observations, ensuring documentation while allowing flexibility (Royal

Academy of Engineering, n.d.). Following the installation, short follow-up questions were asked to clarify participants' understanding of the process, why certain choices were made and their perception of a successfully implemented speaker system. The full design of the observation material is presented in Appendix A.

After installation, the participants used the speaker system in their everyday environment for a month. The purpose of this extended phase was to capture user behaviour, evolving perception and frictions. The participants were asked for continuous reflection and feedback by documenting thoughts, experiences and notable events. Inspired by the principles of Cultural Probes (Gaver et al., 1999), the documentation materials were designed and personalised to stimulate engagement and thoughtful responses, rather than structured forms. This was done to ensure answers without the presence from the researchers. Lastly, when the observation material was retrieved an informal interview was held to collect final thoughts from the participants of the full user experience.

## 3.5. Qualitative Interviews: Semi-Structured

The qualitative interviews conducted in this study were semi-structured. This interview format allowed for discussion around the topic, while still providing the flexibility to explore unanticipated themes as they emerged (Newcomer et al., 2015). However, semi-structured interviews are time consuming and unpredictable. A solid interview guide was therefore crucial and the guidelines presented in the *Handbook of Practical Program Evaluation* (Newcomer et al., 2015) were used as a foundation.

The interview guide began with a few, introductory questions to help set a positive tone. While the question in the interview guide aimed to follow a logical flow, it was expected that the conversation might deviate from the guide. To ensure that all key areas were covered, essential questions were highlighted as a priority item. The guide consisted primarily of open-ended questions, followed by "Why" and "How", which encouraged more detailed responses. Everyday language was used throughout to provide clarity and ease the understanding for the interviewees. The full interview guide can be found in Appendix B.

## 3.6. Brainstorming

Brainstorming was performed as a method for idea generation. The method was selected as it supports critical and creative thinking and enables the exploration of multiple alternative solutions rather than focusing on a pre-defined concept (Norman, 2013; Converse Willkomm, 2018).

The brainstorming process was initially conducted individually to encourage independent idea generation and reduce group bias. Thereafter, the individually generated ideas were discussed in a group setting, where they were evaluated, built upon and combined to enhance their overall quality and feasibility (IDEOU, 2025). Questioning each other's ideas constituted an important part of this process, even when questions appeared obvious or trivial. As Norman (2013) emphasises:

"The obvious often is not obvious at all. What we assume to be obvious is simply the way things have always been done [...]"

This reflective approach encouraged the reassessment of implicit assumptions and supported the development process. The iterative brainstorming session was complemented with summarizing potential shapes and features in a moodboard.

### 3.7. Workshop for Application Interface Design

The workshop was conducted as a method for evaluating early prototypes of the application's layout. By testing the prototype with its intended users, valuable insights could be obtained into how users actually interact with the product (Norman, 2013).

Expected outputs and outcomes were defined prior to the workshop, in order to design activities that supported the intended results (Stickdorn et al. 2018). The outputs were communicated to the participants as deliverables. In contrast, the outcomes concerned analytical insights, formulated as design related questions to be answered through participant interaction. For example, one intended outcome was to gain insight into which interface elements should remain permanent within the application.

According to Stickdorn et al. (2018) it is important to create a safe space for participants in co-creating settings. In this workshop, a safe space was established through informal conversation at the beginning of the session, followed by a short introduction round in which all participants had the opportunity to present themselves. This approach aimed to foster familiarity among participants and to ensure that everyone felt acknowledged and comfortable expressing their perspectives.

## 4. Pre-Study

In the pre-study chapter, Marshall is examined with focus on its brand identity, design language and product portfolio. This analysis is based on a combination of desk research, observation, interviews and informal interactions with personnel at Marshall. These methods form the foundation for a comprehensive understanding of the brand. Subsequently, a competitive analysis is conducted to assess Marshall's position within the market and to evaluate how the company competes in relation to its key competitors.

### 4.1. The Brand Identity

Marshall heritage lays the foundation of the brand's identity, originating from the early stages of its history which are still visual today. As the Director of Integrated Marketing describes the company in three words

*"Raw, Relentless & Unapologetic"*

- Director of Integrated Marketing

It is evident that Marshall's brand identity is deeply rooted in its long lasting cultural and aesthetic legacy. The goal from the brands perspective is to be a conversation starter for music enthusiasts that claims music as a vital part of their everyday life. Marshall wants to express something beyond their product features. The brand aims to create an emotional connection with consumers by associating with music, artistic expression and personal identity. Their product should also serve as a means of expressing thoughts and feelings that might otherwise remain unspoken.

## 4.2. The Design Language

Marshall has a distinctive design language that is recognisable across all their products, whether they are speakers or headphones. Originally the cabinets were wrapped in leather, however they have switched to a vegan water based PU leather instead. To keep the heritage and iconic feel of the portable and robust speakers, which are wrapped in a silicon sleeve, this material is made to mimic the texture of the leather whilst enabling water and dust resistance.

They have three prominent colours that are found on the majority of their products. The three main colours are Black, Creme and Brown, all of which are visible in Figure 3. In addition some products are also available in the colours Midnight Blue and Burgundy. When doing collaborations with artists or other brands, the ability to change colourways and some materials are possible but the overall body and shape of the product remains the same.



*Figure 3: The three standard colours of the Marshall portfolio, Black, Creme and Brown*

Marshall has preserved the original shape of its early amplifiers, defined by rectangular enclosures with rounded corners. Regardless of variation of size or function these box-like shapes are consistently found throughout the speaker portfolio.

The Marshall logo plays a central role in the visual identity of all their speakers, placed on the front of the product to emphasise the brand recognition. Most commonly the logo is made out of brass, variations in white or black also appear across different models.

The control panel, which is made out of brass, looks the same on the majority of the speakers and includes different knobs, buttons and the powerswitch, with rubber detailing see Figure 4. To indicate the status of each control, LED-lights are integrated into the panel to for example indicate the current volume level. The knobs do not feature an indication line on the upper surface. The design choice is due to the speaker also being controlled through a mobile device. The digital LED-lights therefore align with both physical adjustment as well as steering digitally.




*Figure 4: Brass control panel with knobs, buttons and powerswitch (Marshall, n.d)*

However, some of the portable speakers are equipped with a rubberised control panel where buttons are integrated into the top of the speaker to indicate different features, see Figure 5. This design highlights the speakers portability, underscoring robustness as well as water and dust resistance. The LED-light bars on these models are multifunctional and adjust the indication according to the feature being used at the given moment, for example changing volume or adjusting treble.




*Figure 5: Rubberised control panel of the portable speaker Middleton II( Marshall ,n.d)*

The membrane piping, typically located on the front of the speakers, is made out of a mesh material. Depending on generation and type of speaker, this mesh may consist of paper fret, woven metal or fabric. Some of the products come with a handle, either made by silicone, fabric or a leather sleeve with a red or taupe velvet inside, see Figure 6.



*Figure 6: Velvet inside of the handle of Stockwell (Marshall, n.d)*

## 4.3. Portfolio

Marshall's portfolio consists of robust and tactile speakers which originate from the amplifiers that were created by the brand in the 60s. Today the speaker portfolio covers Home, Portables, TV Sound and Party speakers. Marshall offers a unique form of multi-directional sound, True Stereophonic, which is exclusively integrated into the majority of the speakers. The speakers all create a broad spectrum which delivers high quality sound while still keeping the retro feeling of the manual knobs and visual aspects of the design.

All of the existing speakers that Marshall sells today are supplied with the next generation Bluetooth solution, Auracast. This allows for the receiving of the Auracast signals which Heddon and Heston, the latest products, can transmit. The Bluetooth connectivity has a range of 10 meters, but using the Auracast function instead increases this range to up to 25 meters making connectivity throughout a house possible with the Heddon at a centred position.

A feature that is often integrated into the speakers is the tactile control panel, including knobs and buttons. This allows for turning on/off the speaker, play, pause and skip tracks as well as customization of the bass and treble on the speakers. One button unique for Marshall speakers, is their M-button, which provides easy access to presets, which may involve specific combinations of volume, bass and treble settings. Furthermore, the M-button serves an additional function by being utilised in the establishment of the Auracast connection.

During this project the main focus and inspiration was taken from the portable and home speaker portfolio.

### 4.3.1. Home Speakers

The home speaker group is built up by Acton, Stanmore and Woburn, all of which are in the third generation, see Figure 7. (Marshall, n.d.)




*Figure 7: From left; Acton, Stanmore and Woburn (Marshall, n.d.)*

The home speaker line-up are all AC-Powered and connect wirelessly to your phone through Bluetooth or with an input by cable to any source that supports RCA or 3.5 mm.

This category is complemented with the music streaming hub Heddon, see figure 8, that connects to multiple speakers, playing a synchronized sound throughout one's home. (Marshall, n.d).




*Figure 8: Heddon (Marshall, n.d.)*

The Heddon is built upon the Auracast technique which both enhances the reach of the music and allows for as many speakers as one desires to play together. The interface issue is solved by communicating with Heddon through Wi-Fi. By a one time set-up in the Marshall App, the Heddon connects to the Wi-Fi and by simply opening Heddon's connection state and pushing buttons on the desired speakers, all speakers will listen to the broadcast. After creating a group, if one speaker wishes to be disconnected from the larger group, the speaker's Bluetooth button can overrule the synchronised state and simply connect to another music source (eg. phone). The Heddon also allows for older versions of speakers to be connected by RCA cable, therefore Heddon is both backwards and forwards compatible. The RCA input on the Heddon can on top of this allow for more analogue sources to be used, such as Vinyl players or similar, bridging the gap between analogue richness of vinyl and modern streaming.

### 4.3.2. Portables

To be able to reach the customers who are on-the-move, Marshall has developed 6 different types of portable speakers. In order of magnitude the smallest is Willen, then Emberton, Middleton, Stockwell, Kilburn and the biggest is Tufton. All of which have some kind of handle to ease the use and transport between different locations (Marshall, n.d). All portable speakers are found in Figure 9.



*Figure 9: From the left; Willen, Emberton, Middleton, Stockwell, Kilburn and Tufton (Marshall, n.d)*

The Willen, Emberton and Middleton are all Waterproof and have been tested to withstand 1 meter of water for 30 min. The Stockwell, Kilburn and Tufton are water resistant and can withstand some water splashes, rain and dust. The playtime ranges from 17+hours in the Willen, 20+ hours in the Stockwell and Tufton, and up to 30+ hours in Emberton and Middleton. The longest playtime is found in the Kilburn that allows for more than 50 hours of active playtime.

The Marshall signum of tactile control knobs are found in the Stockwell, Kilburn and Tufton, allowing for control of volume, bass and treble.

### 4.3.3. TV Sound

Marshall has recently launched a TV Sound solution in the shape of two soundbars, Heston 120 mm and 60 mm, as well as a subwoofer, Heston Sub 200, see Figure 10. The soundbars have the technology of the Heddon built into it which allows for wireless connection between the soundbars and subwoofer. The subwoofer can therefore easily be put anywhere in the room to enhance the base of the TV and supply a wider range of sound to the TV experience.



*Figure 10: From the left; Heston 120, Heston 60 and Heston Sub 200 (Marshall, n.d.)*

### 4.3.4. Party

The party line-up consists of the Bromley 750, see Figure 11. This portable large speaker allows for 40+ hours of active playtime and has been developed with a removable battery allowing for easy change to increase the time even more (Marshall, n.d).




*Figure 11: Bromley 750 (Marshall, n.d.)*

The Bromley 750 is designed to create a portable comprehensive music experience including inputs for microphones, dj-setup, instruments as well as having integrated stage lights into the front. It has a luggage handle and wheels which makes transporting of the weight of 23.9 kg easier for the user.

## 4.4. Competitive Analysis

The brands examined in this analysis were Sonos, Bose, JBL and Bang & Olufsen (B&O). Initially Harman Kardon was also included in the competitive analysis. However, during the data collection, their products were not found at the selected resellers. To ensure equal comparability across all brands in the analysis, a decision was made to exclude Harman Kardon. Data was collected through unannounced, short structured interviews with sales personnel in the stores. To avoid potential biases from the sales personnel, the introduction was made as representatives of KTH instead of Marshall.

To summarize all collected data from online research and interviews with retail sales personnel, each company was described through concise responses to three questions: *What characterises each brand?, Frequent pains,* and *Frequent gains*. The resulting summary is presented in table 1.

*Table 1: Resulting summary of competitive analysis*

| Brand | What characterises each brand? | Frequent pains | Frequent gains |
|---|---|---|---|
| Marshall | Design language<br>Sound quality<br>Tactile (volume, base, treble) | Faulty items delivered<br>Battery quality reduces<br>Lack smart features | Robust<br>Powerbank feature |
| Sonos | Closed ecosystem<br>Subtle design<br>App | Connectivity issues<br>Unintuitive | Water and humid resistant (works in bathrooms) |

| | | | |
|---|---|---|---|
| | | Software issues (sudden stops, changes volume, non accessible app) | High quality feeling |
| Bose | Portable<br>Powerful bass<br>Robust | Not fulfill promised battery time<br>Expensive<br>Low volume | Sound quality<br>Smart speaker (voice control) |
| JBL | Robust<br>Compact (great sound to size)<br>Portable | Equalizer only accessible through app<br>Instable connection<br>Battery quality (active playtime) reduces quickly | Water and heat resistant<br>Good base (compared to size) |
| B&O | High quality feeling<br>Design language<br>Luxurious feel | Unreliable connection<br>Portable speakers lower sound quality than stationary ones<br>App issues<br>Price | Stereo<br>Water and dust proof |

Thereafter, brand mapping was conducted for Marshall and their competitors. It is important to highlight that the resulting position is a reflection based on the outcome of the competitive analysis rather than how the individual brands would like to be perceived and position themselves. The brand mapping is visualized in Figure 12.




*Figure 12: Brand mapping based on the competitive analysis*

Several of the categories align with how Marshall positions themselves. For example, Marshall is strongly positioned towards a statement piece. A number of reviews highlighted the iconic retro design as something positive, which correspond with the brand identity Marshall wants to express. Furthermore, Figure 12 indicates that Marshall and B&O are the primary competitors in this category.

However, other categories reveal divergence between Marshall's self perception and how the brand is perceived by customers. One noticeable category that diverged were whether customers are casual listeners or passionate listeners. Marshall describes their customers as highly passionate music listeners. In contrast, retail sales personnel explained Marshall's customers, often as younger, who enter the stores already familiar with the brand, usually through friends, looking for a recognisable speaker.

Also a noticeable category is whether their speakers are considered to be used as stand alone or in an ecosystem along with several speakers. In Figure 12, Marshall is positioned strongly towards the stand alone category. In Figure 12 one can also see that Sonos is alone competing in the category of having an ecosystem. With Marshall's newly launched Heddon, this will leverage their position in this category, moving more towards ecosystem, meaning they can compete against Sonos in this category.

# 5. User Study

In the following chapter, the results of the user study are presented, which includes the survey, observations and interviews

## 5.1. Surveys

The surveys were distributed across different online platforms, resulting in 90 answers. Since it was unknown how many people the survey reached, an answer rate could not be determined.

According to the survey, Bluetooth speakers are the most common choice among younger users, largely due to its simple setup, straightforward connectivity method, lower cost and portability. In contrast, respondents who use Wi-Fi speakers most often rely on Sonos, typically chosen for its multiroom functionality.

Across all respondents, sound quality emerged as one of the strongest determinants when selecting a speaker. However, ease of installation also plays a significant role in the decision making process. While many users are price sensitive, they still express a clear desire for smart features and care about design. The most common concerns regarding new technology relate to privacy and data usage, as well as the risk of malfunctions or technical problems.

One hypothesis that was investigated was whether there is an association between speaker type and reported problems. Based on the collected data from the survey, it could be determined that there exists an association between speaker type and reported problems. Figure 13 displays the reported problems, in percentages, from each category: Bluetooth, Wi-Fi and a combination of Bluetooth and Wi-Fi. A clear difference in reported issues between users of Bluetooth speakers and users of Wi-Fi speakers was observed. From the group of Bluetooth users approximately 20% reported problems, while 70% of Wi-fi users reported problems.

The Chi-squared test resulted in a p-value of 0.00089. Given that this value is below the 0.05 significance threshold, the null hypothesis of no association between speaker type and reported problems could be rejected (Zaveri, 2024). This indicates that there is a clear statistical relationship between the variables. Thus, the data provides supporting evidence for the hypothesis which states that users of Wi-Fi speakers report problems more frequently than those of Bluetooth speakers.

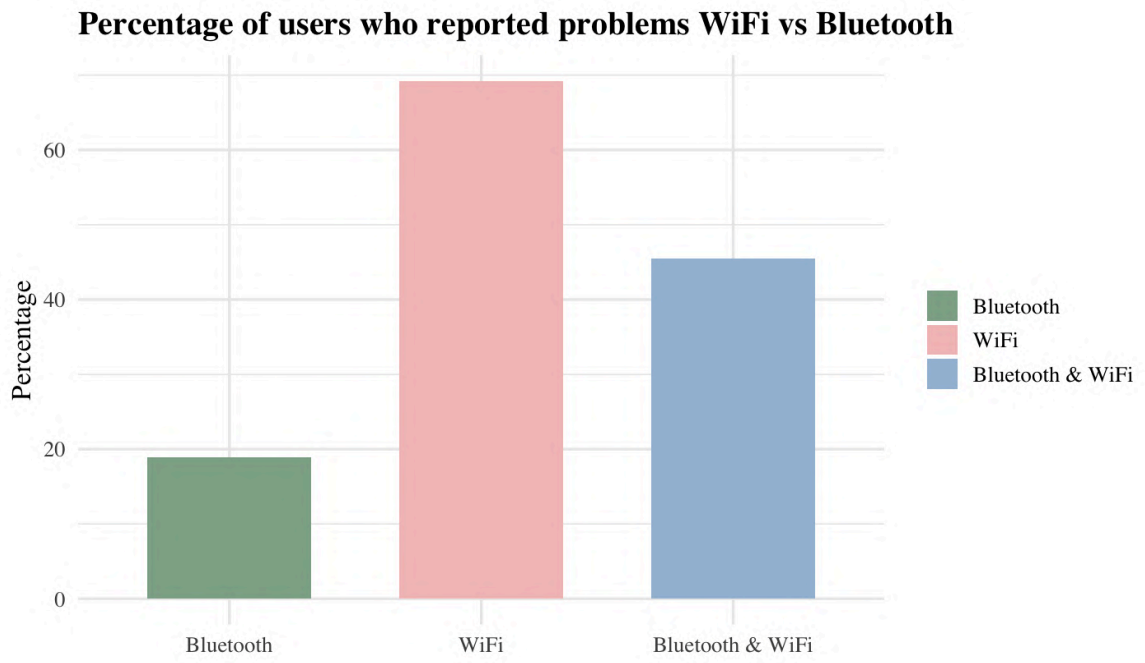



*Figure 13: Plot showing the relationship between amount of problems encountered with different types of speakers*

Further, Figure 14 illustrates how different levels of technological confidence correspond to the types of speakers respondents use. A clear pattern emerges, Bluetooth speakers dominate among respondents with lower technological confidence. Reflecting that the users tend to prefer devices perceived as easy to set-up, reliable and low maintenance. This aligns with the commonly reported motivation of simplicity as a reason to buy Bluetooth speakers.

Conversely, respondents reporting the highest level of technological confidence have a more diverse distribution of speaker types. These systems often require more advanced configuration, integration with networks or use of companion applications. The broader acceptance of such systems among high technical confidence users indicates that they may be more comfortable troubleshooting connectivity issues or navigate more complex installations.

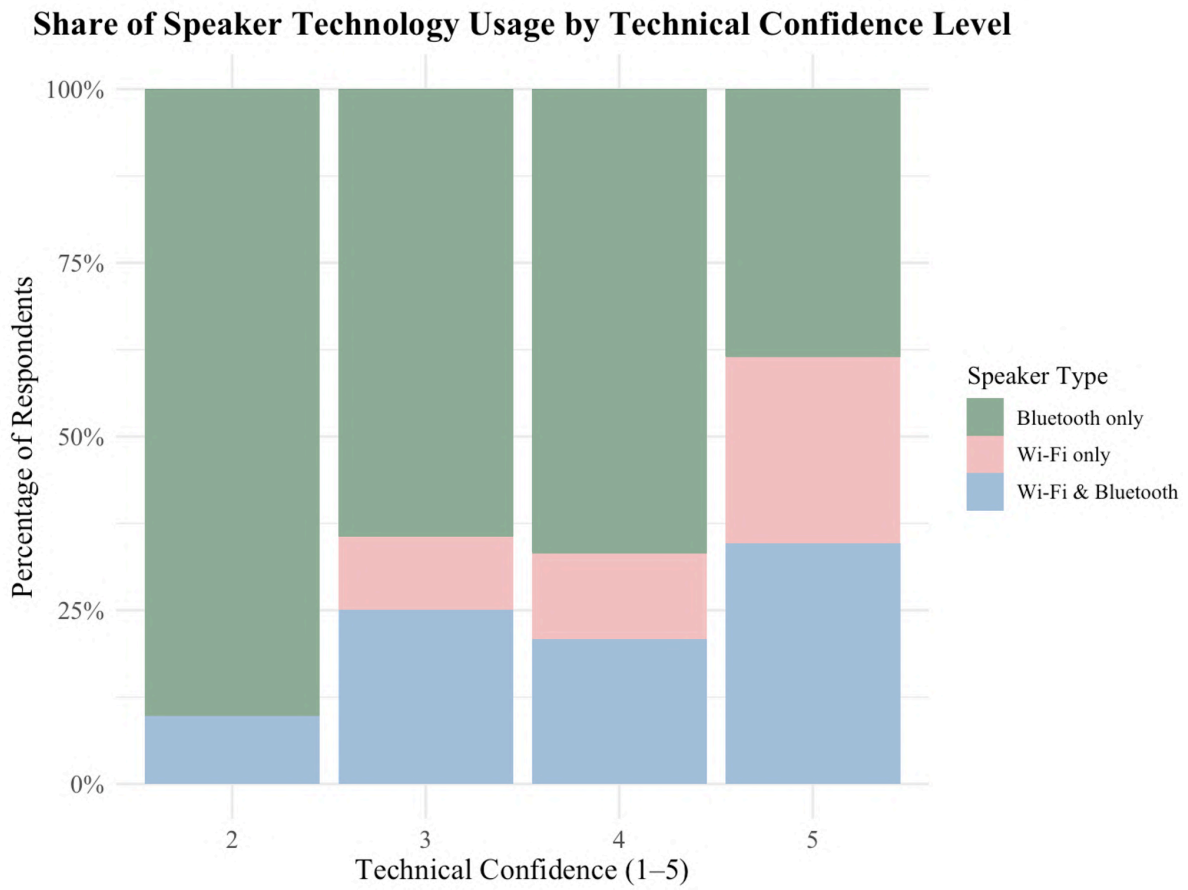



*Figure 14: Plot showing the type of speaker type owned across levels of technical confidence*

## 5.2. Observations

During the two observations the installment processes were monitored. One observation was conducted in a domestic setting, while the other took place in an office environment. The observation in the home setting was intended to examine user case I, involving multiple speakers requiring synchronisation, and user case III, where a single speaker is used across different locations. The office environment observation primarily investigated user case I, as well as user case II, which concerns multiple users wishing to play music together. In reality, both observations were relevant to all user cases, however, each observation primarily focused on the two cases described above.

### 5.2.1. Observation in Domestic Setting

Observation one was conducted in a multiple leveled house where two people live, see Figure 15. In the house there were a total of five speakers and one Heddon, during the installment. Yet only four were used during the whole trial month.

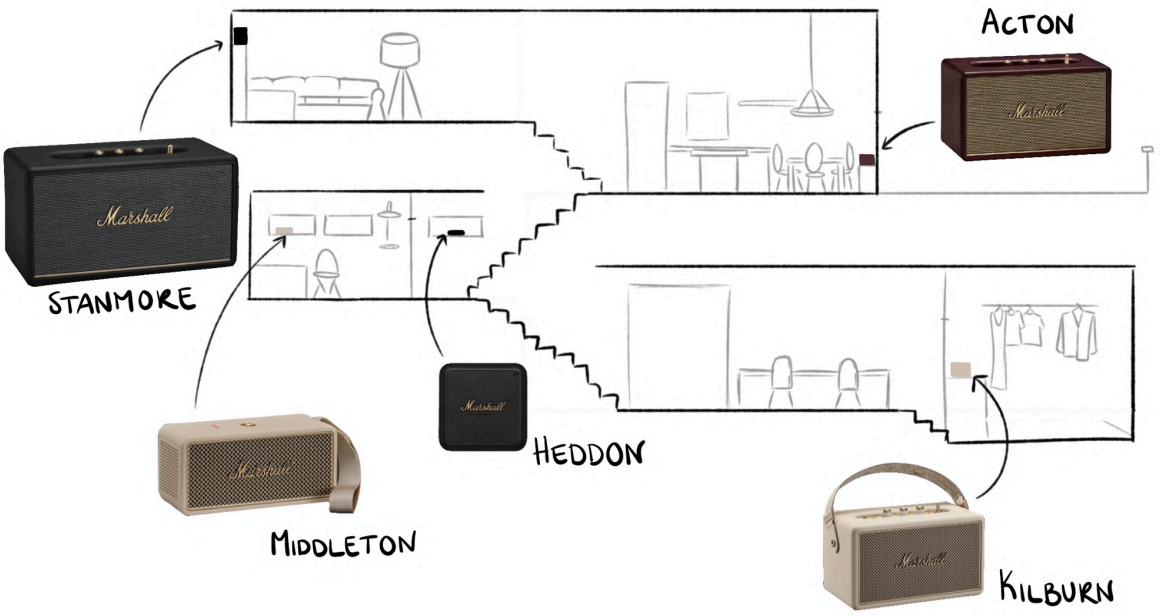



*Figure 15: Floorplan of the house and placement of the speakers and Heddon*

The installation started with going through the different products where the testers were given information that they would normally get from the store. They were supplied with three portable speakers, one Emberton, one Middleton and one Kilburn, and two stationary speakers, one Stanmore and one Acton. They were also supplied with a Heddon. The overall information of features and functions were then briefly discussed, such as what the different speakers were communicating through their design. Based on these incentives, placements of the different speakers were proposed by the participants. Middleton and Emberton communicated water resistance and more robust design, therefore one of them was proposed to be placed in the bathroom. The Kilburn speaker, though portable, did not communicate the same on-the-go design. One of the first acknowledgements when the speakers were presented, was the resemblance between Acton and Kilburn. The participants did not see any difference in them and had a hard time identifying if they had different functions, although Kilburn has a handle and should communicate portability. The stationery speakers as well as the Kilburn, all of which are supplied with knobs and buttons in a similar set-up, were immediately responded to with a positive attitude. Both participants started to play with the knobs and stated that they liked the bigger buttons and tactile feel of them. During the installation of Emberton and Middleton, the play button was hard to interpret and pressed several times in a frenetic way.

Thereafter, the installation process began, conducted from one participant Iphone. The Heddon and all the speakers were installed and connected in the kitchen area at first, then moved to their designated locations.

Two apps were downloaded for the installation process, which caused confusion regarding why multiple apps were necessary. Both participants expressed a preference for a single app. Throughout the installation process further problems were encountered. For example, the application interface was presented in English while the operating system of the phone was set to Swedish, which caused confusion due to the mixing of languages. Furthermore, having english

configured as the app language created additional difficulties, particularly in relation to speaker-specific terminology. The option to select or change language during the setup was not presented, and the option to change later on in the application was perceived as difficult to locate.

When attempting to locate an overview of all installed units, it was observed that the participants repeatedly pressed the homescreen button despite already being on the home. This behavior indicated that the interface structure was unclear and some labels and navigational elements appeared to be missing. See Figure 16 to see the layout of the current homescreen within the application, as well as the interface displayed when the plus icon is selected, enabling navigation between all installed speakers.

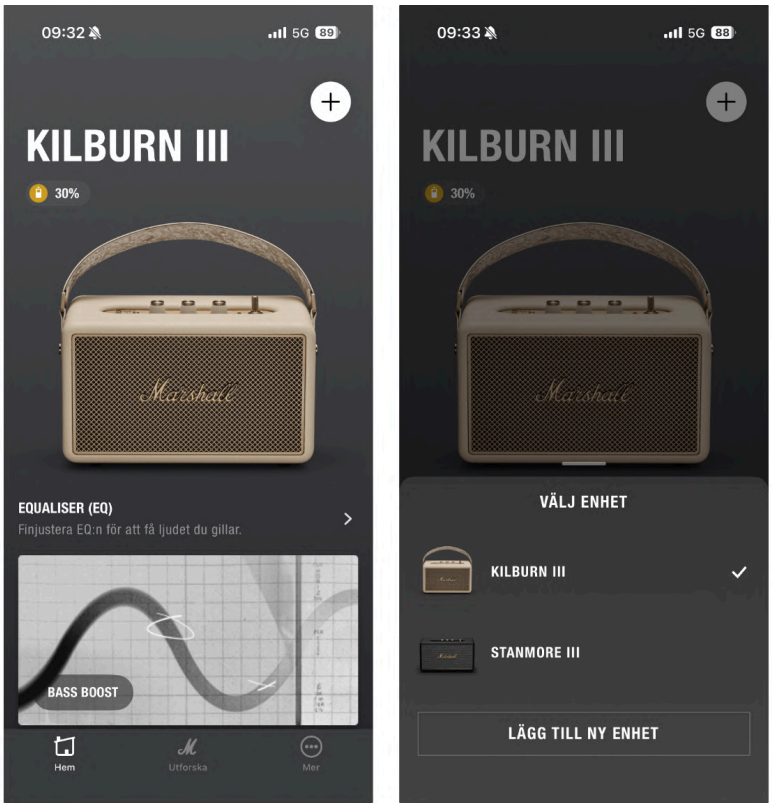



*Figure 16: Homepage and all installed speakers when pressing the plus icon on homescreen*

Additionally, during the step-by-step guided installations of individual speakers, some steps lacked clear current state feedback. For instance, when establishing the Bluetooth connection between the phone and the device, the process involved a loading period. However, the loading indicator was both small and visually diffuse, left unnoticed by the participants, it contributed to uncertainty, see Figure 17.

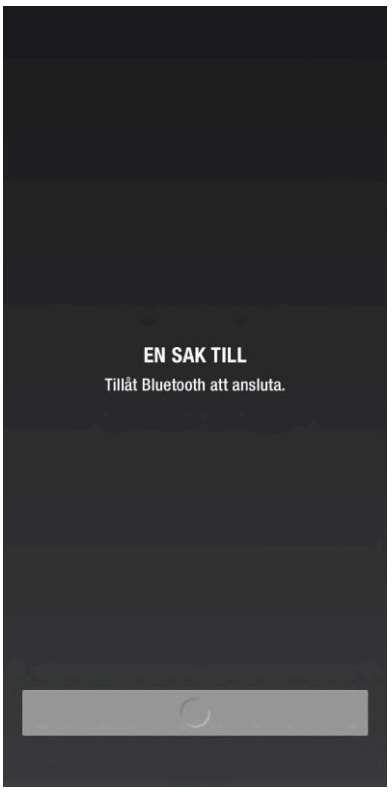



*Figure 17: Loading page during Bluetooth connection between phone and speaker*

This prompted participants to attempt to proceed to the next step prematurely, actively searching for a "next" button and expressing confusion about the system's current state. Further confusion arose through a linked button which redirected users from the Marshall app to the phone's settings. The link led to an incorrect page in the settings app, where Bluetooth already appeared to be enabled. Participants were redirected to a page granting the Marshall app permission to use Bluetooth, rather than to the Bluetooth configuration overview. This increased uncertainty about which action was required.

Another unclear moment occurred due to the combination of Bluetooth and Wi-Fi connectivity technologies within the setup process. When connecting the Heddon, participants attempted to locate it within the Bluetooth menu of the phone's settings, despite the device operating over a Wi-Fi connection.

An additional source of uncertainty emerged in regards to Bluetooth connectivity, as the setup process required a dual configuration involving Bluetooth Connection and a Bluetooth LE connection. The initial connection involved pairing the speaker with the phone and was carried out by the participant. Subsequently, an additional connection was required to establish direct communication between the speaker and the Marshall application, which was automatically initiated by the application itself. This distinction was not made clear within the application interface, the question arose and was therefore explained by the observers during the session. The system itself did not supply the users with sufficient information at the time of setup.

Due to the time required to install multiple units, the portable speakers were turned off after a period of time. This led to confusion regarding their status, as it was unclear whether the speakers were powered on or off.

After the installation was completed, one participant who uses a hearing aid connected to the phone via Bluetooth received a phone call. Under normal circumstances, the ringtone is heard directly through the hearing aid, and the call audio is routed to the ear. However, with the Emberton speaker present in the house, the speaker instead became the primary audio output when the call was answered, preventing the call from being heard through the hearing aid. It was not possible to switch the active Bluetooth connection back to the hearing aid. Different actions were tried to solve the issue without success. To prevent further disturbance, the Emberton speaker was therefore removed and excluded from the remaining observation time.

An important consideration is that the participants in the household shared a single Spotify account. As a result, music playback activity may appear as originating from a single user, despite involvement from two individuals. However, this aligns with this observation's primary focus on user cases I and III, rather than user case II, which involved the scenario of multiple users playing music from the same speaker.

During the test period, each participant received a sealed letter continuously throughout the period, containing reflective questions relating to their experience with the speaker system. Participants were instructed to answer the questions individually and return the responses back in the envelope.

By the end of the test period, the letters with written reflections were collected, and a concluding informal interview was conducted to gather insights into the user's experience.
Overall feedback was positive and the system will be missed once the speakers are retrieved. Both participants stated that they appreciated the design of all speakers and found the features easy to navigate. The visual, audio and haptic responses from the speakers were highlighted as valuable, as they enhanced usability and contributed to a smoother user experience. Playing music throughout the house had become part of their daily routine.

Music playback was initiated through Spotify and streamed to the Heddon. The stationary speakers began playing immediately, while the portable speakers were powered off, as they automatically shut down after a period of inactivity. As the user moved through the house during their morning routine, the portable speakers were powered on and automatically connected to the speaker system, resuming playback of the same music. Manually activating individual speakers was not perceived as problematic, as reconnection to the system functioned seamlessly.

During the one month test period, the users encountered one bigger issue. The stationary speakers began to make a crackling sound and subsequently failed to play music, despite both Heddon and the speakers indicating by light that they were active. An observer was contacted, and the issue was resolved by restarting the entire system, including unplugging the Heddon. After this event, the participants expressed a desire for clearer indicators showing whether the

speakers were connected to the Heddon, either through visual feedback on the speakers, within the app, or both.

Another issue raised concerned incoming phone calls while music was playing at a high volume. When answering a call, lowering the speaker volume through the phones was difficult, as the volume controls adjusted the call volume rather than the speakers, which was perceived as inconvenient.

### 5.2.2. Observation in Office Environment

The second observation was conducted at an office where eight people work, see Figure 18. During the installation, three participants were present. This observation also provided insights into how effectively the installers could communicate the functioning of the systems to others at the office.

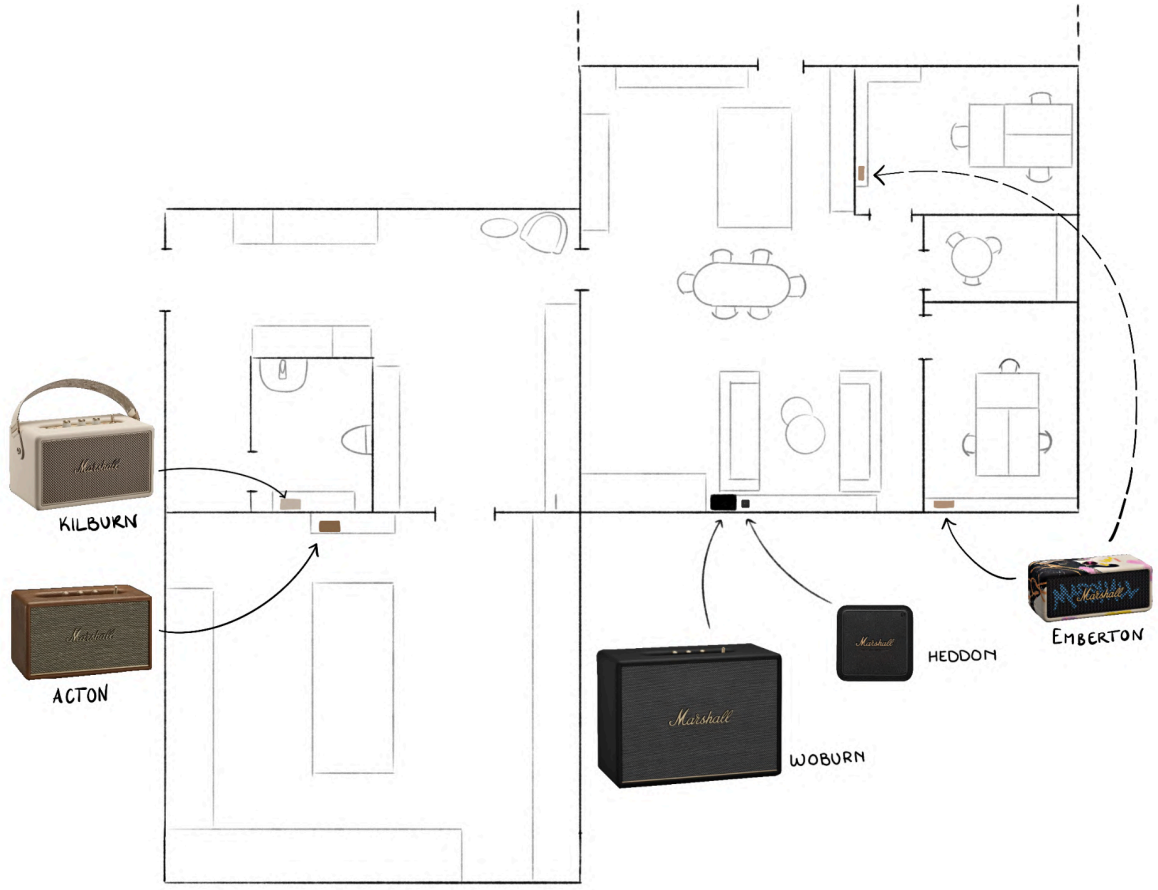



*Figure 18: Floorplan of the office and placement of the speakers and Heddon*

Heddon was placed alongside Woburn in the central lounge area of the office. Acton was placed in the warehouse while Kilburn was installed in the restroom. Emberton was positioned in one of the office rooms, but during the test period it was moved between the two main office rooms. The installation plan included one speaker for each office room, however due to logistic constraints, only a single speaker was available to serve both offices.

Similarly to the installation in the domestic setting, the office observation started with going through the different products where the participants were given information that they would normally get from the store. They were supplied with three portable speakers, one Emberton, one Middleton and one Kilburn, and two stationary speakers, one Woburn and one Acton. They were also supplied with a Heddon and the overall information of features and functions were then briefly discussed.

In the office environment, participants similarly directly expressed positive attitudes regarding the manual, physical controls present. However, the buttons on the Middleton and Emberton were perceived as harder to read and interpret. Participants appeared to place greater emphasis on audio, haptic and visual feedback from the speakers than the ones in the domestic setting. These sound cues were described as both confirming and supportive, especially given that some symbols and texts were smaller and provided limited visual feedback.

The installation began with downloading the needed applications, as before the need to have two separate apps were not met with positive feedback. The Heddon was firstly connected to the app which presented minor challenges. When the Heddon was being installed, the instructions guided the user to press the button until pulsating light appeared. Before pressing on any button, the light already pulsated, and the participants assumed the Heddon was already in the open state even though it was not. The observers were forced to intervene and explain that the button needed to be pushed even though the already pulsating light. The users pressed the button until a new pulsating rhythm appeared, though very similar to the first. This caused confusion amongst the participants whether this step was necessary or not. Further, after installing Heddon when following the connection steps  instruction in the application the participants were automatically redirected to an external webpage intended to guide the pairing of the speakers. This transition came with a noticeable change in layout, typography, and colour scheme. Participants interpreted this page as a “terms of use”, hence the textheavy layout, and therefore exited it, returning to the application without noticing nor completing the required installation steps, see Figure 19. At this point, the observers were forced to intervene again, explaining the function of the webpage, even after clarification the page was seen as text heavy, with small typography that made it difficult to read.

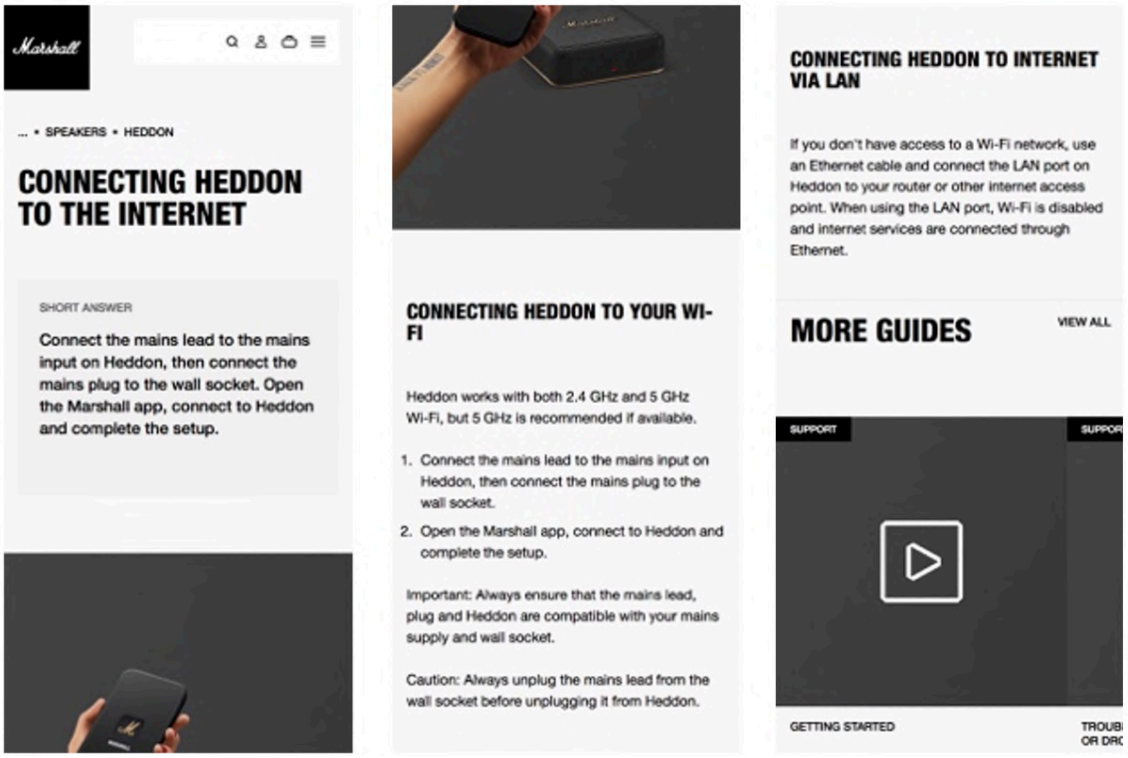



*Figure 19: Instructions on how to install Heddon when directed to an external website*

During the installation, as in the domestic setting, participants expressed confusion regarding the need for two separate Bluetooth connections. Without the presence of the observers, participants stated that they would not have understood this dual connectivity and likely would have assumed that an error had occurred. During the first connection to a speaker, to confirm that the connection had been successful, participants proceeded to Spotify directly and started to play music. This method was attempted rather than checking the connection status in the application or the Bluetooth menu on the phone.

As the installation process proceeded, the names assigned to the speakers were seen as difficult to read and remember. Communication between the participants were somewhat confusing at times due to small lettering, however the visual consistency between the speakers as presented in the application and their physical appearance was described as both helpful and reassuring. This facilitated recognition and orientation within the interface. Additionally, based on the different model names as well as the different locations of all speakers the participant expressed a need for a clear and accessible way to rename the speakers within the application. Several attempts were made to locate such a function but without success.

After each speaker was installed, pop-up windows appeared containing information about functions and features for each specific unit. The participants immediately dismissed these as an expressed desire to firstly verify the connection as successful before engaging in additional information. This sequence contributed to a perception that relevant information was not presented at an appropriate time during the process.

Once all units were connected, participants noted the absence of a home screen providing an overview and statuses of all units within the app.

To ensure usage of the speaker system, a playlist was prepared in advance and shared with all employees at the office. This measure was intended to lower the barrier to playing music, as the process of choosing music may otherwise discourage use. One participant used the Spotify playlist to start playing music in the system and the other participants joined in by creating a Spotify Jam.

In the office setting, a post box was installed, see Figure 20, allowing participants to anonymously leave notes about their experience. To lower the threshold for contribution, supporting questions were placed next to the box.

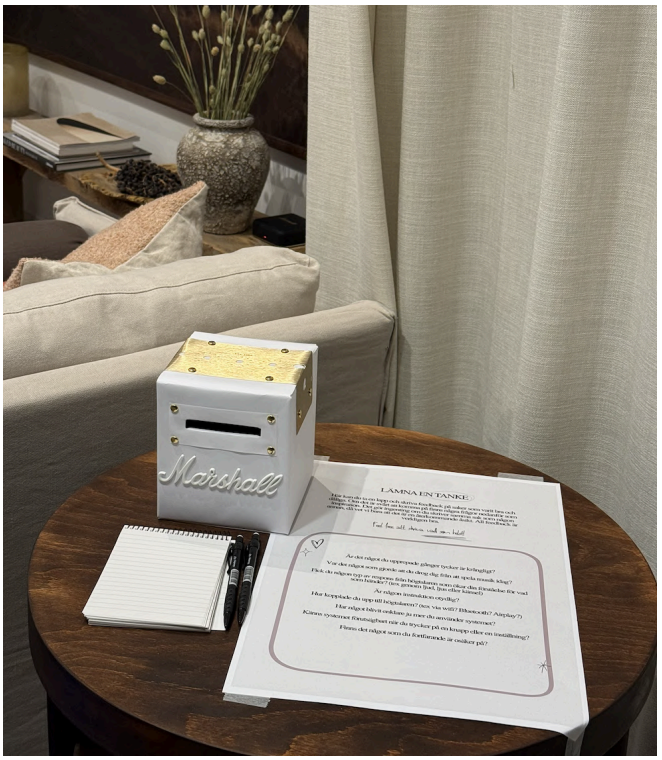



*Figure 20: Letterbox, paper, pen and accompanying questions*

Additionally, a visual board listing common daily usage scenarios were installed together with a phone on a door in a passageway, see Figure 21. Participants could mark and comment occurrences daily on the board, providing event-based data as well as a reminder to reflect on the user experience.



*Figure 21: Thoughtsheet and accompanying pen placed in passageway*

After the test period the speakers and materials were collected and during which an informal interview was held to get information of the overall experience. During this interview, a user who was not present during the installment process, expressed uncertainty of the system and the initial playback had never been activated by her.

Similarly as in the domestic setting, the speaker system was well appreciated and will be missed. The great sound and bass was compared with the existing Harman Kardon speaker and Marshall was seen as superior. It was noted that, in some instances, the music kept playing after the individual who initiated playback had left the premises, whereas in other cases, the music stopped once this individual exited the building. The different speaker behaviors observed were attributed due to inconsistent connectivity methods across different users in the office. Questions regarding when music should continue and stop playing were raised during these events. These issues suggest that users may not fully comprehend the underlying technique of the speakers and that essential information was not communicated sufficiently.

### 5.2.3. Common Interface-Related Observations

This chapter presents a summary of the interface design findings from both observations, which were largely similar. The main problems regarding the external design were identified in the interface of the Emberton, see Figure 22, and the Middleton, see Figure 23. Overall, the observations revealed several issues related to the interpretation of physical controls, visual indicators and feedback mechanisms across both devices. Multiple participants actively searched for a distinct “play” symbol, missing on both speakers, indicating the need and expectations of clearly labeled control panels.

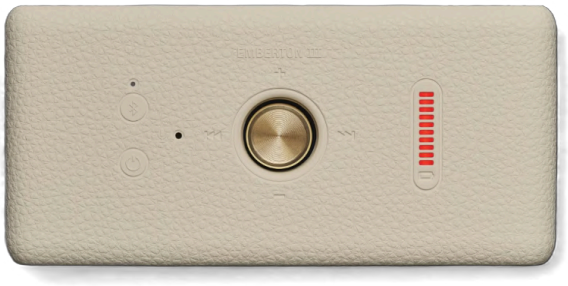


*Figure 22: Interface of the Emberton (Marshall, n.d.)*

On the control surface of the Emberton, the placement and functions of symbols were frequently misinterpreted by the participants. The symbols for volume and track change buttons are placed outside of a smooth silicone ring around the jog. Several of the participants pressed directly on the flat textured silicone surface, such as the plus symbol, instead of interacting with the jog control as intended. In addition, participants attempted to press the battery symbol, suggesting uncertainty regarding which elements were interactive and which were visual helpers. Distinguission between the Bluetooth button, the power button as well as seeing the name and microphone symbol on the speaker was challenging for all participants. The microphone symbols on both speakers were not acknowledged by any participants and the LED was therefore seen as a status lamp for the actual speaker.



*Figure 23: Interface of the Middleton (Marshall, n.d.)*

Similar issues related to the control panel were observed in regards to the Middleton, although additional complexity arose from the light-based feedback. The participants failed to understand that the LED-bar on the left was connected to multiple functions, such as volume, bass and treble. When using either of these functions the LED-bar indicates the level related to each function, which was not perceived by any of the participants. Furthermore, participants struggled to recognise that volume adjustments were represented by three subtle brightness levels before shifting to a higher step. The differences in light intensity were perceived as too diffuse, even after explanation by the observers. As with the Emberton, the power and Bluetooth button as well as the name of the speaker was hard to see. Before attaching the handle, some participants mistook the mounting as a button, though unclear of which purpose. Additionally, the LED indicator for the microphone was misinterpreted as referring to the power status.

Across both speakers, the jog control emerged as a problematic interaction element, seen as the circular brass detail centrally placed on both speakers in Figure 22 and 23. Participants described

the jog feeling "spongy", lacking both clear directional guidance and tactile feedback. The combination of both weak haptic response and unclear light signals, led to beliefs that nothing was happening when pressing the jog, this in turn caused frustration and irritation. Several participants assumed the jog were to be pressed to emerge from the surface, allowing it to be turned rather than used as a button. Misinterpretations such as these made it difficult for participants to follow instructions such as “press the play button”, searching for a separate, clearly defined button. To conclude, the jog received predominantly negative feedback, with many participants suggesting that it should be redesigned or replaced with more intuitive physical buttons.

Further, the different models of speakers used in the observations were supplied with a variation of power buttons. On the Stanmore III, the power button has a clear switch feature, where the stick shifts from an up or down position. The Kilburn III, on the other hand, has a rebound effect on the stick, always returning to the top position regardless of being on or off. Participants held varying opinions about which button was the preferred one, though it was evident that when the buttons were used in combination, confusion emerged.

## 5.3. Interviews

In total a number of 8 interviews were conducted, both online and in person. The interviewees were between the ages 24 to 68, with different technical skills, interests and living situations. The selection of interviewees are visualized in Figure 24. This selection of the interviewees were made in order to gain a broader spectrum of insights that would generate the greatest of valuable information.

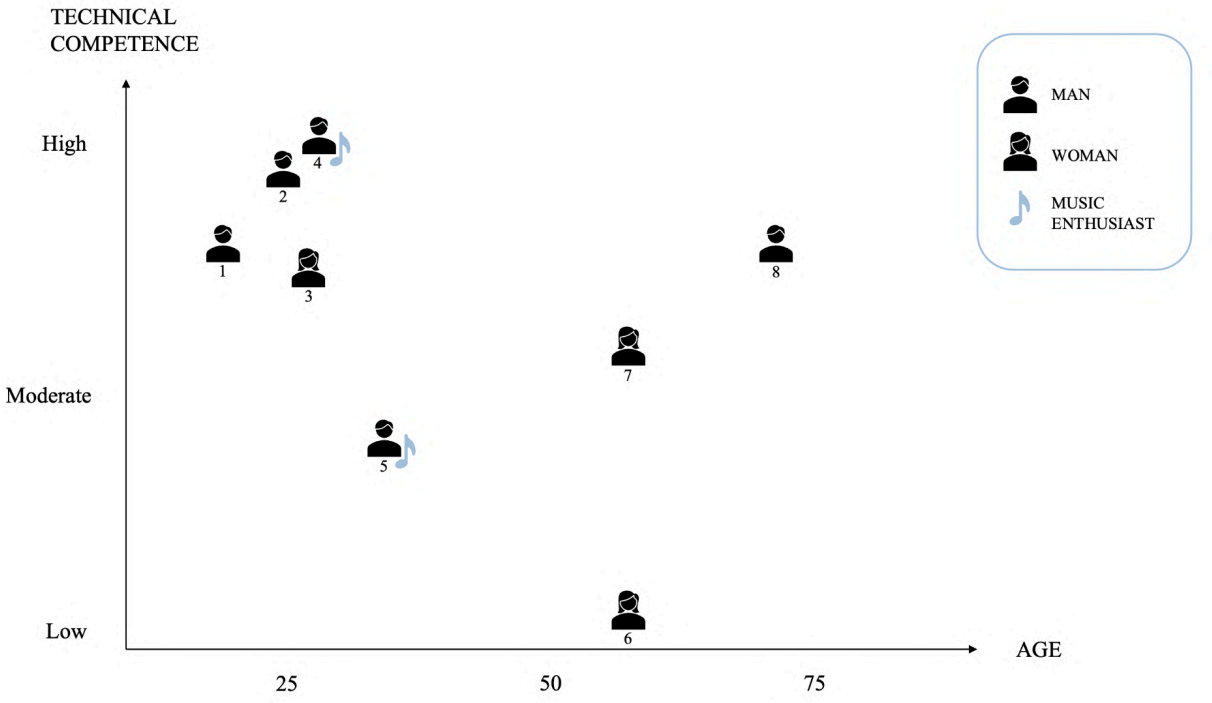



*Figure 24: Visualisation of interviewees*

Interviewee six and seven are Android users, while the rest of the interviewees are IOS users. Interviewee one and three have small portable speakers from JBL that are rarely used either inside or outside the home. This was due to lack of sound quality and low battery life. Interviewees two, four and eight have Sonos speakers. Number two and four chose them due to familiarity as they had Sonos speakers when they grew up. They had one or two speakers in total. While interviewee eight had an ecosystem consisting of seven Sonos speakers and chose them due to its leading multiroom function back when they were bought. Interview six was given an old Harman Kardon, which is mainly used during social gatherings. Interviewee seven have an ecosystem consisting of Google speakers compatible with her Google home and Android phone. Users tended to either choose speakers due low price or sound quality, overlooking one or the other based on interest.

Some of the key insights regarding the technology of the speaker were irritation regarding installation where loss of connection occurred or reinstallation was needed. A frequently mentioned need was when a device returns to the home environment, such as after a visit to the park or a trip, the device should automatically reconnect to the system. The same applies when the speaker dies, when fully charged again and powered on it should automatically reconnect. This was seen as a threshold to use the speakers, leaving fully functional speakers untouched.

*"It keeps losing connection every time it dies or is brought outside the home, and I have to reinstall it every time, so I basically stopped using it."*

- Interviewee 2

It was noted that several interviewees did not feel the need to understand the technology behind an action but more importantly that the technology worked and the steps towards an end-goal was understood.

The interviewees liked step-by-step guides, where actions were explained and guided through the accompanying app. Interviewees commented that they liked when the steps were linked to the action they had to perform. For example, when the app provided the prompt "Connect your device through Bluetooth settings" the word setting was highlighted as an interactive element. Tapping it would lead the user directly to the Bluetooth menu in the device's settings.

*"I like when the buttons are highlighted in blue, so I know I can press on it and it will lead me forward to perform the action."*

- Interviewee 6

However, a recurring problem mentioned was when the step-by-step guide deviated from reality. In some cases, a button or a menu item referenced in the instructions did not exist in reality, leaving users unable to proceed and without any alternative path and forcing them to restart the process. Such discrepancies may stem from outdated app-versions, device specific variations or software faults.

The users expressed a negative attitude towards app-dependencies, even though they acknowledge that such applications may be necessary. Apps occupy storage space, which many users currently experience limitations with.

*"A lot of stuff requires apps these days and it is nice not to need an app because they take up so much space on your phone."*

- Interviewee 3

If several people wanted to control the music, participants reported using Spotify Jam, which was regarded as the conventional method for shared music control. No interviewee experienced any issues with this method.

New technology was not actively sought out by the interviewees, even those with substantial technical knowledge and interest. Instead new technology was only adopted if it came with an added value, and even then the younger interviewees were reluctant if the price was considered high. It is important to note that all interviewees emphasised the need for technology to be tested before they were willing to invest time and money into it. Tutorials, videos, and other forms of guidance were frequently used as complementary resources when assistance with the product or technology was needed. Consequently, it was considered essential that such support materials were available before purchase.

*"I do not get easily fascinated by new technology [...] I'd really like there to exist guides and info before I invest my time into it."*

- Interviewee 4

The interviewees were asked to connect and play music to their speaker and observations were conducted to examine the different ways. The two most common ways were through Spotify's own connecting tool, Spotify Connect, or through Apple's Airplay. Participants selected their preferred method based on which option they perceived as the fastest and most reliable for establishing the connection. All of the interviews expressed confusion regarding the different connectivity methods and the advantages associated with each. A frequent pain point brought up by the interviewees regarding Bluetooth speakers was the interruption of music by notifications and other sounds from phone activity from the pairing device. Such interruptions caused confusion of the source of the sound, as well as irritation among individuals within the proximity of the speaker.

When talking with the interviewees regarding features of portable speakers some of the most desired characteristics were suitable size, low weight and high durability, attributes that all express the affordance of portability. The expressed need related more towards movability than portability in the traditional sense. For instance, moving the speaker between areas within the household was seen as more important rather than taking it along when leaving the home.

If the interviewees were to bring the speaker outside, a durable battery time was preferred. A big reason why portable speakers were not used outside the home today was the lack of battery or

not being fully charged when needing it. Loss of charging cables was also a recurring problem and a desire for quick-charging was expressed.

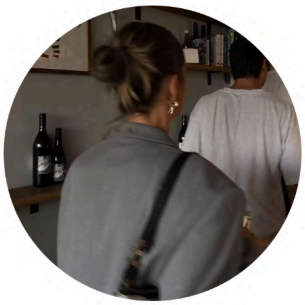
Some of the indications were mentioned as more relevant than others, for example connectivity sounds and battery-level indication lights were expressed as important for multiple interviewees. Recognisable symbols and standardisations were also highly appreciated, especially if the technology were unfamiliar, as such cues supported user understanding and helped adopt the new technology.

Lastly, speakers today were generally seen as a technology product rather than design objects. While products such as Sonos and Google Home are intentionally designed to blend seamlessly into the interior, more distinctly "design-oriented" speakers were not dismissed. The offerings today were regarded as lacking the technology capabilities users excerpt. Given the balance between price and technical performance, the interviewees were not willing to choose a more aesthetically unique speaker at the expense of functionality.

## 5.4. Personas

Based on insights from the survey, observation and interviews two personas were designed to represent potential customers. A template for creating each persona was found at Openlabs website (Openlab, n.d.). Persona one, Saga, is presented in Figure 25 and persona 2, Mikael, is presented in Figure 26. Each persona is described by their goals and motivations, desires, wishes or ideal experience, frustrations, behaviours and other important details.

Persona one, Saga, is a 30 year old who lives with her boyfriend. She values creating a warm and cohesive atmosphere in her home, particularly during social occasions such as dinner gatherings, where music plays an important role. Music contributes to the welcoming and aesthetically unified environment she aims to establish across different rooms. Her boyfriend typically handles troubleshooting of technical products and supports her when connectivity issues occur. A recurring barrier in her everyday use is that the speaker is often uncharged, partly because locating the charger is inconvenient and the charging process is perceived as an unnecessary effort. Due to her dyslexia, she is less inclined to read instructions or engage with text-heavy interfaces. As a result, she prefers products with intuitive design, require minimal explanation and can be quickly understood without extensive reading.



**SAGA**

30 Years old

Two-room apartment with balcony in Göteborg

Lives with her boyfriend Samir and their cat Leya

Works as HR at Thelins

*"WHEN SOMETHING DOES NOT WORK I TEND TO NOT USE IT"*

**GOALS & MOTIVATIONS**

She likes her job

She loves to host dinners for her girlfriends at her apartment

She is interested in clothes but is driven by trends, wants to set her own style

Shares a lot on social media

**DESIRES**

Play music whilst hosting dinners

Would like music in the bathroom

Be able to bring the music out to the balcony

Don´t want to think about charging units

Samir likes a great bas whilst Saga thinks more about the esthetics

**WISHES / IDEAL EXPERIENCE**

A speaker that fits into the home but does not blend in too much, a first step into her own style

Helps Saga in her routine that she has established

**FRUSTRATIONS**

Speakers are not fully charged

The charger is gone

Issues with connectivity

**BEHAVIORS**

On the weekends they eat breakfast outside on the balcony while listening to a podcast together

Enjoys go for walks

**OTHER IMPORTANT DETAILS**

Planer

Lacks good patience

Dyslectic

If something does not work Samir is the one who takes over

*Figure 25: Persona 1 - Saga*

Persona two, Mikael, is a 45 year old divorcee with two children. When alone, he enjoys listening to all kinds of music at a high volume, however he values his classic genres and artists. A problem he often experiences is a decrease in sound quality when the volume increases. Together with his children, they enjoy exploring music together, which frequently leads to discussions about preferred levels of bass and treble during different songs.

Mikael feels comfortable with keeping up with the latest technology, but when technical issues arise he tends to blame the product rather than reflecting on his own usage. The new technology needs to be established and add value before he decides to invest in it. If reviews and user guides exist for the product the threshold is decreased. Mikael's home consists of multiple conversation starter pieces, and he wished for his technology to serve a similar role, something that encourages discussions with friends and family.



**MIKAEL**

45 years old

Divorced

Two Children, Sam 10 & Leo 13

Project leader at a start-up

Stockholm

*"I DO NOT GET EASILY FASCINATED BY NEW TECH, I WANT TO KNOW THAT IT WORKS BEFORE I INVEST TO MUCH INTO IT"*

**GOALS & MOTIVATIONS**

Highly focused by his job

Likes to go out and socialise with friends over a glas of beer

Likes hockey

Likes to cook food

Likes to stay up with the latest tech

**DESIRES**

Listening to music whilst cooking

Enjoys a deep bas

Likes to feel young

Likes to show of his skills

**WISHES / IDEAL EXPERIENCE**

Want´s to have conversation starter pieces in his home

Does not want to be tied up to one single brand

**FRUSTRATIONS**

The volume is too low sometimes when he is vibing with the music

The children mixes with the settings on the speaker

**BEHAVIORS**

Pancake breaky with the boys on Saturdays whilst listening to music together on high volume

**OTHER IMPORTANT DETAILS**

Accuses the technique before analysing his own behaviors

Stuck in his own routines and ways of working

Favorite Artist: Bruce Springsteen

*Figure 26: Persona 2 - Mikael*

A storyboard, found in Appendix C, was made based on persona number one, Saga, to enable identification of events and behaviors that could occur during a day of speaker usage. Some of the key pains worth mentioning was the unclear visualisation of battery life, both during use and while charging. Further, interruptions when playing music due to notifications on the phone created a disruption of the ambiance. Lastly, automatic reconnection did not work seamlessly which caused frustration.

On the other hand, the identified positive experiences were firstly the enhanced overall impression of having music playing over multiple rooms. Secondly, Spotify Jam was used without friction as a default method when multiple people wished to play music together. Thirdly, the ability to adjust volume, play and pause, both on the phone as well as on the speaker, were seen as valuable features.

# 6. Findings

In the following chapter, the key findings derived from both the pre-study and the user study are presented and discussed. The chapter aims to provide a cohesive understanding of the results and to identify insights that informed the design decisions.

Grounded upon the research, the applications emerged as a key area for improvement, with the potential to enhance the user experience and create added value. Furthermore, the research indicated an opportunity to develop a movable speaker that supports synchronisation between speakers. A summary of the findings from the pre-study and user study are presented in Table 2 below. The findings are structured into observed behaviours, then conclusions drawn from these

behaviours, and corresponding insights that could be considered in the final delivery. The insights implemented into the final concept are highlighted in bold.

*Table 2: Summarising table of behaviours and identified findings and insights*

| BEHAVIOUR | FINDING | INSIGHTS |
|---|---|---|
| COMPETETIVE POSITIONING | | |
| Marshall positioned towards statement piece along with B&O | Goes hand in hand with how Marshall describes themselves | **Important with their iconic design to keep competitive advantage in this category** |
| Marshalls customers described as casual listeners looking for familiarity | Divergence between how Marshall describes themselves and how general public describes them | Purchase motivation may be driven by brand familiarity rather than quality |
| Sonos competes almost alone in category Ecosystem | Sonos have created generic trademark in with their ecosystem, which makes it hard to compete against them in this category | **Important to educate customers with functions Marshall products have, without scaring them away** |
| SPEAKER | | |
| 83% choose BT speakers | Comfortable and familiarity with technique | **Simple & recognisable set-up, Bluetooth should be kept as an option** |
| 69% problems with Wi-Fi | Blames speakers instead of units | **BT should always be available as an option** |
| Speakers are chosen often based on price | Stands in relation to sound quality and design | Can't develop an too expensive speaker |
| Customers prioritise ease of installation | Methods with recognisable elements ease the installation | Time to music should be short and intuitive |
| Observation participants interpret design features of the speakers | Mentioning features such as handles and material | **Communicate movability and stability through design, both through shape, material and features** |
| Observation participants positively comments on knobs and buttons | Likes tactile feeling of bigger knobs and buttons | **Interactions should be through intuitive and tactile elements** |
| Resemblance between products causes confusion | Visualise differentiating functions | Communicate distinction between products |

| BEHAVIOUR | FINDING | INSIGHTS |
|---|---|---|
| Participants did not push pairing due to already pulsating light on Heddon | Pulsating light of Heddon already pulsated when asked to enable pairing | Visual, audio and haptic responses should be differentiated to not be too similar in regards to different functions |
| Portable speakers found existing system when being turned on again after battery saving mode | Automatic reconnection is preferred | **Include long-term memory in products which allow for automatic reconnection** |
| Participants pressed on the silicone around the playbutton and did not interpret the LED lights behaviours on the Emberton & Middleton | The play button was hard to understand and use due to the placement of the symbols in combination with the silicone material it made it hard to visually see the different symbols | **Clear visualisation of symbols by contrast and intuitive placement , enhancing understanding with visual, haptic and audio feedback/response align and work together to create reassuring feedback for user** |
| Expressed insecurity when pressing volume, bass and treble on Middleton, since the level of volume was not visually perceived | Did not interpret the behaviour of the LED list to the action performed | **Multiple functions in one element can cause confusion and take away rather than ad value, hence buttons should be kept to one function** |
| Users did not understand if the portable speaker was on or off due to the similarity of power buttons when used in combination with the acton | Confusion regarding status when similar elements have inconsistent design or behaviors | Differentiate between functions |
| Interviewees expressed frustration regarding charging. Did not know how much battery the unit had, visually see if the speaker was charging and frustrating of locating the charger | Hard to know when the unit is charged, what battery life is left and if it needs charging | **Allow for fast charging and ease charging experience. Communicate battery life and charging status with indicators** |
| Do not wish to be too careful and think about the speaker too much | The speaker should feel sturdy and be easily moved around | **Communicate sturdiness and movability** |
| Irritation when the music is paused when receiving | The Bluetooth connection does not allow for playback | Either use Wi-Fi or communicate why events |

| BEHAVIOUR | FINDING | INSIGHTS |
|---|---|---|
| phonecalls | | happen |
| Users did not express a need to use speakers outside of the home as a priority | The need is more towards moving a speaker around the home | Allow for movability |
| Users choose Marshall if friends and family have them | Choose on brand familiarity | **Keep the iconic and recognisable design of the products and tactile parts** |
| APPLICATION | | |
| Multiple apps are commented as a frustration | Difficulties with identifying when and where to locate features and functions | **One app for everything** |
| The ability to change language in app was not found during observations | Objects and functions in application are hard to locate | **Intuitive core structure/flow of application** |
| Observation participants missed loading phase and prematurely looked for next-button | Diffuse visual indicators in installation guide | **Clear status feedback of installation process and make contrast between elements clear** |
| Home symbol is pressed multiple times by observation participants | Overview of all connected devices is missing | **Homescreen or similar that visualises all devices and connectivity** |
| Played music to check if speaker was successfully connected | Connectivity status of speakers after installation was not interpreted | **Clear visualisation of connectivity status, on speaker and/or in application** |
| Linked buttons in installation guide led to wrong page in phone settings which caused confusion | Confused users more than it assisted them. | **Enable linked steps to the right pages** |
| Participants searched for the Heddon in BT settings | Users don't understand the difference between BT and Wi-Fi as connectivity methods | Application needs to communicate differences |
| Participants thought the LE BT connection was an error | Confusion and missing information in application | Communicate clearly what is happening during installation process |
| Stress and confusion when receiving phone calls while | Hard to adjust the volume of the speaker system when in | **Allow for easy access of master volume to adjust** |

| BEHAVIOUR | FINDING | INSIGHTS |
|---|---|---|
| listening to music | an active phone call due to the volume of the call being controlled | **the volume of the whole system or Bluetooth which pauses the music.** |
| Pressed away the Heddon connection set-up steps located in external webpage | Thought the webpage was a terms-of-use due to switch of design layout and text | **Consistent design throughout user experience, in application and when moving to other external Marshall pages** |
| Participants could visually connect application with physical products due to aligning picture | Reassurance of coherence between reality and digital world | **Align reality with digital application** |
| User got stuck if step-by-step guide deviated from reality and had to start over | Frustration when step-by-step guide is not aligned with reality | **Troubleshooting guide needs to be easily accessible during all steps of the installation process** |
| Participants expressed difficulties remembering product names of the different devices | Referred to products with own names | **Possibility to rename the products with own names** |
| Pop-up windows of the installed unit containing information about the product was pressed away by users | Relevant information should be accessible at the right time, users wants to know that the speaker works before doing anything else | **Time to music should not contain added steps, treat them as added value that can be accessed within the application when the users needs it** |
| Frustrations regarding being tied up to an app | Users understand the need to have an app at installation but does not wish to use it further | The application should add value rather than be a compulsory tool |
| | TECHNOLOGY | |
| No reflection of why a certain connectivity method is used | The different playback methods are not understood by the users, the user simply use the method they been taught | Clearly communicate pros and cons of different connectivity methods |
| Interviewees expressed reinsurance when online reviews and guides exist before investing in new | Not interested in new technology if it is not tested and troubleshooted before, bugs should be worked out | Make sure to add troubleshooting and help support, maybe do videos of usage or likely |

| BEHAVIOUR | FINDING | INSIGHTS |
|---|---|---|
| products | | |
| Expressed no real interests in technology behind a feature but cared more about the added value or end goal | Flow and steps toward the end goal are more important than educating users | Communicate why an action is needed towards the end-goal but not the technology behind it |
| Users play music together by using Spotify Jam | Spotify Jam is seen as the accepted way | **No new techniques for joint music playing is needed** |
| Users tend to be confused when prompted with new technologies | Hard to time motivate investment if not clearly communicated added value | **Market them as a feature rather than the technology** |

## 6.1. Pre-Study Findings

Based on the competitive analysis, several insights related to strategic positioning were identified. Marshall was strongly positioned towards a statement piece alongside B&O. It is therefore essential that Marshall continues to deliver speakers with their recognisable design and tactile feeling in order to maintain their competitive advantages in this category.

Further, retail sales personnel described the typical Marshall customer as younger, already familiar with the brand, rather than customers who are music enthusiasts, as Marshall describes their customer. This indicates that customers' purchase motivations, at least in Sweden, may be driven more by brand familiarity rather than looking for the highest performing speaker, which is likely to be a more important criterion for passionate music listeners.

Moreover, findings from the field study indicated that Sonos currently holds an exceptionally strong position in the ecosystem segment. According to one sales representative, a significant number of customers enter the store requesting a "Sonos solution" when in reality they are seeking a speaker ecosystem rather than a brand-specific product. This suggests that Sonos has attained characteristics of a generic trademark within this context. However, with Marshalls recently launched Heddon, the company is strengthening its position in this area by moving further towards an ecosystem approach, thereby enabling competition with Sonos within this category. Nevertheless, the observed lack of user awareness regarding alternative solutions highlights a need for improved user education concerning speaker functionalities.

## 6.2. User-study Findings

Firstly, it became evident from both the observations and interviews, that Spotify Jam was the accepted way and worked smoothly when multiple people wanted to play music. Consequently Marshall does not need to prioritise developing an alternative solution for shared music control, at the moment.

Many users report a higher level of comfort when using Bluetooth connectivity. This preference is largely attributed to Bluetooth's simpler setup process and familiarity compared to Wi-Fi, a finding supported by the survey results. Several everyday devices, such as headphones, commonly rely on Bluetooth as the primary connectivity method, which contributes to users familiarity and confidence when using this technology.

Although some users noted drawbacks, such as interruptions caused by phone notifications when using Bluetooth speakers, the technology still offers notable advantages in terms of accessibility and ease of use. The findings indicate that users are more inclined to rely on interaction methods they already recognise and understand rather than trying new ways even though they may come with advantages. Furthermore, when introduced to new technologies or systems, interview participants emphasised the importance of recognisable elements. This could be by integrating familiar symbols and interaction patterns, to facilitate understanding and adaptation.

The survey results further indicate that users of Wi-Fi speakers and systems are predominantly individuals who consider themselves as more technically advanced. This aligns with previous observations that Wi-Fi connectivity is generally perceived as more complex than Bluetooth. According to Marshall, their core customer base does not primarily consist of technology enthusiasts, which supports the alignment between the brand identity and a continued offering of Bluetooth-based solutions. Many users express frustration with complex setup processes and emphasises the importance of fast and seamless system installation.

During the interviews it was made apparent that  step-by-step guides sometimes deviate from reality. Such discrepancies may stem from outdated app-versions, device specific variations or software faults. These findings underscore the need to incorporate alternative pathways within the app, such as alternative methods or troubleshooting options. All of which should be accessible throughout the entire installation. Today's formation forces the user to restart the entire installation process until the desired outcome is fulfilled. This leads to eminent frustration and in some cases results in users abandoning the device altogether.

Further, several steps from the step-by-step guide within the Marshall application were found to redirect users to incorrect pages within the settings interface when linked actions were used. This misalignment caused confusion, as users became uncertain about which action they were expected to perform. Despite this issue, the use of linked keywords and phrases was positively received, both during observations and interviews. The ability to access relevant settings through embedded links were highly appreciated as it supported guidance and reduced navigation effort. It is however of high importance to ensure that links and steps constantly directs users in the intended direction as well as complement with troubleshooting options. Users show a preference for step-by-step guidance with clearly linked transitions between each stage of a process. Further, features such as automatic reconnection are highly valued, as they reduce interaction friction and increase perceived reliability and usability.

Wi-Fi speakers and systems are highly dependent on a stale and continuous network connection. If the connection is disrupted, a speaker that relies solely on Wi-Fi ceases to function. This dependency increases the risk that users attribute technical issues to the speaker itself rather than to the network infrastructure, this may potentially lead to negative product perceptions.

During the observations, participants were able to clearly distinguish between portable and stationary speakers based on their design features. Portability was effectively communicated though differences in materials, form factors and the inclusions of handles on portable speakers. However, similarities between certain speaker models, e.g. Acton and Kilburn, caused uncertainty regarding their respective functions and use cases. This raises the importance of clear differentiation from earlier models when introducing new features that are substantial. If a new product closely resembles previous versions, it becomes difficult to communicate added value and to justify a higher price point, especially when functional differences are not immediately apparent to the user.

Furthermore, when visually similar speakers intended for different use cases were used in combination, such as the portable Kilburn and the stationary Acton, confusion arose due to the identical-looking power buttons. Although the buttons appeared the same, their interaction behaviours differed. The portable speakers button returned to its original position whereas the stationary speakers button remained in a distinct on or off position. This discrepancy in interaction logic led to uncertainty, as users exerted consistent behaviour from visually identical controls.

The differing button behaviours originated from functional requirements, as the portable speaker enters power-saving mode after a period of inactivity which the stationary does not. Nonetheless, distinguishing these controls visually would support clearer affordance and reduce cognitive load. This would enable users to better understand and predict the systems behaviour without having to compare interaction outcomes across devices.

Furthermore, the interview results indicate that users do not consider it important to understand the underlying technology behind products. However, they do express a need to understand steps being performed during interaction at least to a basic extent. An example observed during the study was when a Bluetooth LE connection was being established, a step where users did not understand what was happening during this process. Brief explanatory feedback, such as short descriptions indicating what the system was doing or what the user is waiting for, could have reduced uncertainty.

In line with this, users experienced difficulty identifying the different connectivity methods and playback modes. For example, the observation participants expressed uncertainty regarding whether music playback would continue if the person who initiated the playback left the premises. This behavior suggests that the system did not sufficiently communicate the knowledge needed.

By clearly communicating these distinctions within the application could support users' understanding and decision-making, by making it more apparent when and why each

connectivity or playback method should be used. This would allow users to interact with the system confidently without requiring detailed technical knowledge.

The observer highly appreciated the knobs and buttons on the speaker, which also goes hand in hand with Marshall's identity. Especially liked where the visual, haptic and audio responses, indicating that the user had done it right. However, some responses were similar to each other and hard to notice a difference between and some steps they missed feedback. Differentiating between the responses would help to improve the experience. But overall keeping the manual feel is something that should be done, to continue to appeal to this customer base and not lose the brand identity. This manual feel could also be integrated into the application to clearly weave together these two different spectrums of Marshall. Speakers today are often accompanied by an application which is needed for installation and sometimes further features. While users understand that applications are necessary in the set-up phase, frustrations arise when the application is compulsory to be able to use and interact with the speaker. The application should therefore be seen as a way to add value to the product rather than a compulsory tool.

One observation group missed the opportunity to change language and missed that this could be done in later steps, because they did not find where to do it. This indicated that some value functions in the application were hard to locate. Improving the structure of the application and prioritising what the customer considers as valuable could improve user experience when interacting with the application. When the observation participants pressed the home symbol several times in a row, when already being at the homescreen, further strengthened the argument that the application structure could benefit from an established core structure and flow that matches the customers expectations.

Further, the participants tried to proceed the process when they in reality had to wait for the phone or speaker to synchronize. The application was provided with loading symbols that were missed. This indicated the importance of right size and contrast of symbols to make them noticeable. The same applies to symbols on the actual product, which in some cases, due to their size, contrast and placement caused the observers to perform the wrong action. Such as pressing outside of the button instead of on it. Additionally it was unintuitive that one LED function had several areas of use. Not mixing several functions within the same place is therefore important.

The behaviour of immediately crossing down all information about the speaker after the speaker installation was connected, indicated that this information window was provided to the user at the wrong time. The user wanted to start to play music as soon as the speakers were installed. By providing this information at a time when the user is more available to get this information would increase the value of presenting this information.

Lastly, recurring irritations related to charging were brought up during the interviews. Users reported difficulties to find or interpret the remaining battery level and emphasised the need for clear visualisation during charging. Additionally, several participants expressed a preference for fast charging functionality. Further frustration arose when attempting to locate the charger, as it was often stored away when not in active use. These findings suggest that the charging system

should prioritise both fast and effortless charging in order to improve the overall user experience. Moreover, battery life and charging status should be communicated clearly through indicators, either on the speaker, within the application, or through a combination of both.

An example of one of Marshalls products that could be seen as a difficulty is on Emberton. The battery status bar is temporarily replaced when other features are used, for example, the battery indicator is substituted by the volume indicator during volume adjustment. This was perceived as confusing, as a single indicator was used to represent multiple functions. As previously noted, such dual purpose indicators contributed to user uncertainty. Consequently, the implementation of a dedicated battery level indicator, could be considered.

In summary, the findings from both the pre-study and the user study indicated a clear opportunity to improve the overall user experience, addressing both physical and digital aspects. The insights revealed challenges related to interaction clarity, connectivity and ease of use. Together it motivated the development of both a redesigned speaker and an improved application interface.

# 7. Ideation

The ideation phase consisted of two parts. The first part was brainstorming which were used when generating different speaker concepts. The second phase consisted of a workshop where application layout was ideated.

## 7.1. Brainstorming

Throughout the brainstorming session, different shapes, functions and materials were investigated. Figure 27 shows early sketches of potential shapes of a speaker. Marshall's retro shapes were considered along with more modern appearances. Different docking stations concepts for potential charging solutions were explored in varying forms and configurations.

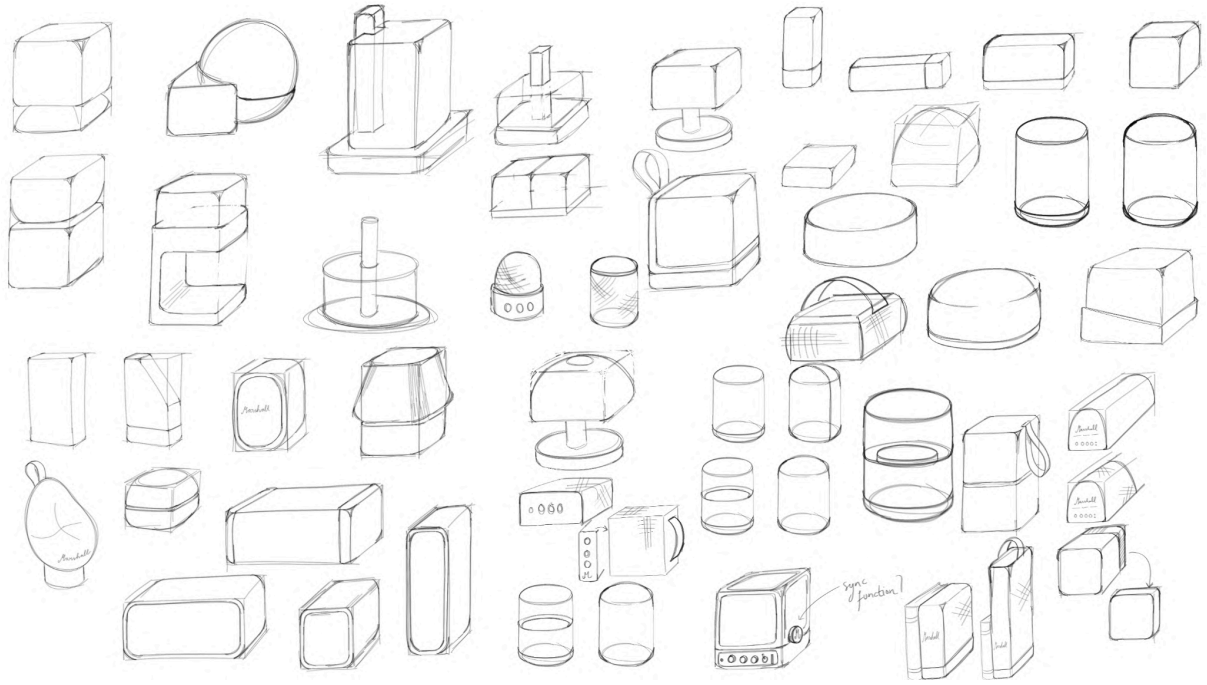


*Figure 27: Brainstorming sketches of shapes and composition*

After the first sketches, different coupling solutions were explored, which are compiled in Figure 28. Varying sizes, levels of complexity and design languages were studied to identify which best align with Marshall's existing design elements.



*Figure 28: Compilation of different coupling solutions*

Thereafter further sketches showed different placement of components, such as the grille and handle positioning. Marshall's retro feel was expressed with the help of different materials and retro shapes, see Figure 29.

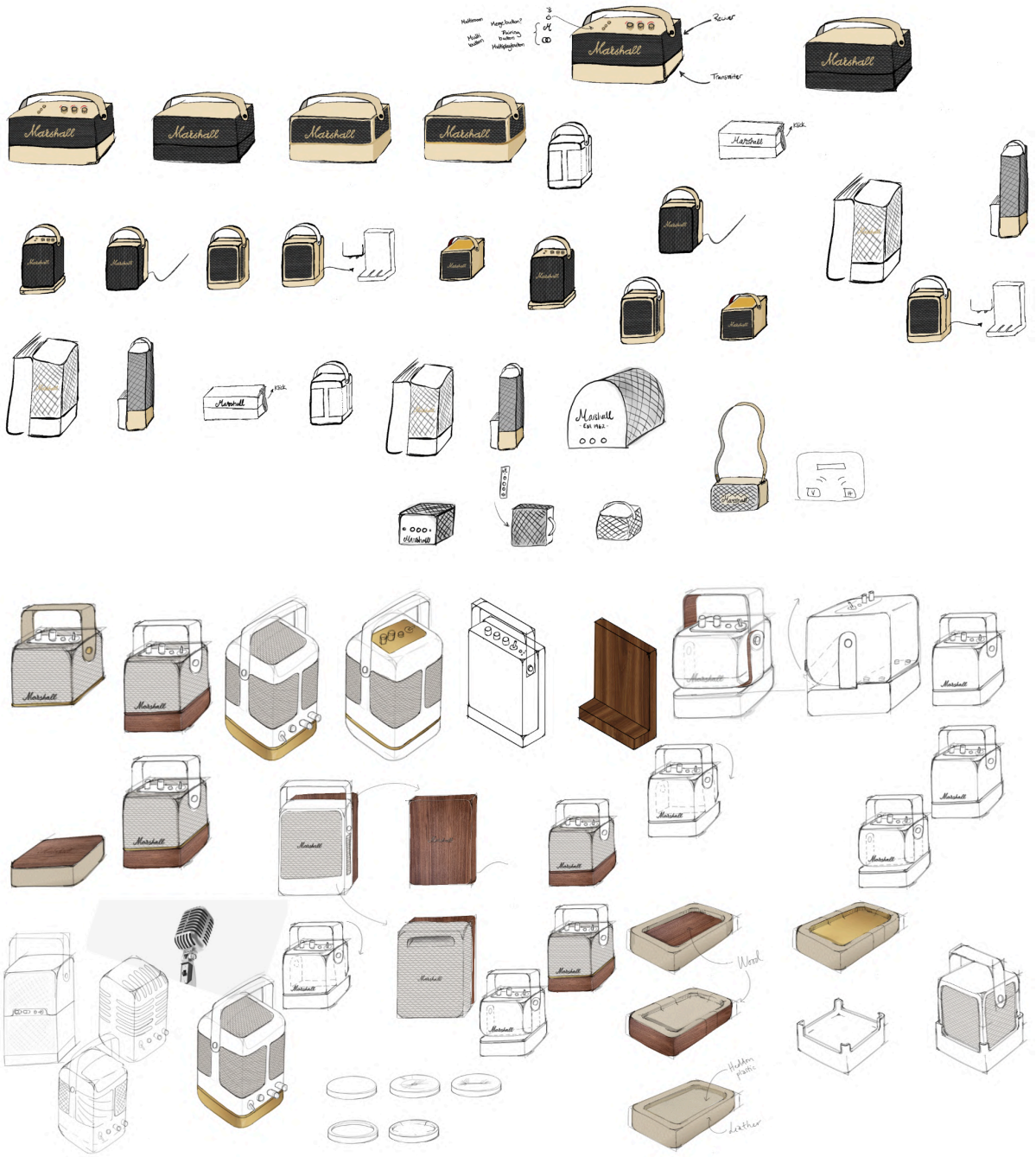



*Figure 29: Early sketches of speakers*

The ideation phase was concluded with a moodboard that articulates the indented aesthetic of the speaker, found in Figure 30. The moodboard conveys a retro inspired feeling through the use of characteristic retro shapes as well as a raw and honest social environment. The intended expression is aiming to appeal to users who appreciate an authentic and relaxed listening experience. The moodboard served as a guiding tool during the later stages of concept generation to maintain focus on intended expression of the speaker.



*Figure 30: Moodboard of the intended emotional and aesthetic expression*

## 7.2. Workshop for Application Interface Design

A workshop was conducted with eight participants representing diverse technical skill levels, occupations and age groups. The workshop began with an introductory round in which participants presented themselves, aiming to establish a comfortable and inclusive social environment. Further, the devices used in the workshop were presented, including their key features and relevant technical aspects, in order to establish a clear foundation for the subsequent tasks.

Thereafter, the participants were divided into two groups and were given a series of assignments. Initially, each group worked independently, after which the results were presented to the other group. During the independent work phase, the workshop leaders circulated among the groups, asking clarifying questions and offering additional perspectives when groups appeared to struggle or needed support in understanding the task. This approach enabled discussions of design decisions, clarification of questions, and identification of both problems areas and essential features.

The first assignment required the groups to design the layout of the applications homescreen. This is a feature that, according to the observations, is currently missing in the existing application. To facilitate this task and make it engaging, the digital interface was translated into an analogue format, see Figure 31. Pre-prepared paper cut-outs representing various interface elements were provided. This along with blank pieces, paper, pens and other materials allowed participants to add missing components if needed. The provided elements consisted of both mandatory components in different formats as well as optional add-ons, thus allowing insights into elements participants prioritised or chose to exclude. The workshop leaders required that all

devices and the connection between them had to be visible in the layout, however, the overall structure and visual organisation were left to each group to define.

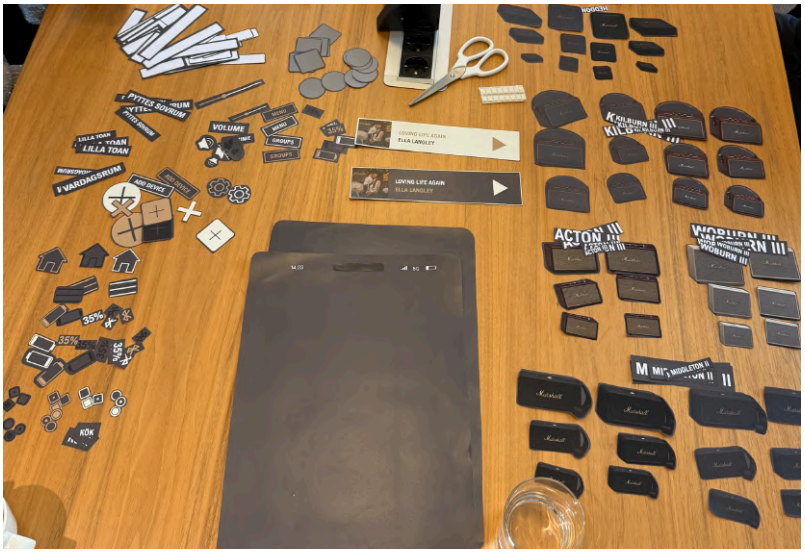



*Figure 31: Workshop material for assignment one, designing the homescreen layout.*

Both groups presented layouts including images of the devices along with personalised names, such as “Livingroom”. This highlighted a shared preference for allowing users to name devices freely, rather than displaying predefined product names. It also became evident that the Heddon needed to be presented differently from the connected devices, reflecting its central “hub” function.

To distinguish between stationary and the portable units, both groups added a battery icon displaying the battery percentage next to portable units. No additional battery-related information was considered necessary. Visual indicators, in the form of boxes, representing the connection to the Heddon were placed near each device. One group proposed that a disconnected device should be manually repositioned to a separate section labelled "Disconnected Devices”. The other group presented the disconnection with leaving the connection indicator unfilled, while keeping the device in the system overview. A small Bluetooth icon appeared to indicate that the device was operated independently via Bluetooth. In both layouts, devices remained visible on the homescreen regardless of their connection status to the Heddon system.

While discussing the homescreen design, the participants also identified interface elements that should remain persistent throughout the application. These were identified as the menu and the home icon. Although the current application includes a bottom navigation bar with the categories *Home*, *Explore* and *More*, this solution was actively excluded in both layouts due to being visually crowded and unintuitive. As a result, both groups favoured a cleaner and more spacious interface. The introduction of a menu was seen as a more intuitive alternative. Instead, a master volume control and music information was placed at the bottom of the homescreen, allowing users to adjust the volume of all speakers simultaneously. Additionally, each speaker was

provided with an individual volume bar, enabling fine-grained control over individual device volumes.

Based on the designed homescreen layouts, the groups were presented with three cases sequentially and asked to adjust their designs accordingly.

In the first scenario, the Acton speaker had lost connection to the Heddon. Both groups displayed a faded visual effect on the speaker, additionally a symbolic indicator was added by one group while the other moved the device to a separate section. In the second scenario, the Kilburn speaker was out of battery. Both groups displayed a battery level of zero percent, and one group additionally dimmed the device visually. In both layouts, the device remained associated with the Heddon. The final scenario involved Heddon losing its Wi-Fi connection. One group presented the Heddon alone in a central position, accompanied by a crossed-out Wi-Fi icon, allowing access to other Bluetooth-enabled devices via a separate Bluetooth button. The other group visualised this scenario by solely dimming the heddon, causing some confusion regarding how to present the other devices, still usable via Bluetooth. Across all scenarios, participants were allowed to modify both layout and functionality of the original homescreen design as needed.

Finally, the groups were asked to design the interface for an individual device page. A key requirement identified by both groups was the clear presentation of the device's connection status to the Heddon. The resulting layouts largely resembled the structure of the current application, though minor adjustments were proposed.

One positively received feature was the incorporation of the physical knobs found on the speakers into the digital interface. While concerns of rotary interaction on a touchscreen, the knobs were nevertheless included as a complementary design element alongside traditional sliders, reinforcing the Marshall brand identity. All finished layouts from the workshop are presented in Appendix D.

The second part of the workshop was conducted individually and focused on exploring how participants interpret different terms and symbols. Since one of the key insights from the user study indicated resistance toward new technology, the Auracast technology, which enables synchronised playback across multiple speakers, needed to be presented as a new function rather than as new technology.

The function of synchronising multiple speakers via Heddon was therefore explained to the participants, who were subsequently asked to describe the function using a single word. In addition, participants were asked to draw a corresponding symbol. Their suggestions were then discussed among all participants.

The most frequently proposed term was different inflected forms of *connect*. Other suggested words were *pair*, *link*, *party* and *sync*. During the group discussion it became apparent that *connect* was primarily associated with linking one device to another, while *pair* was commonly interpreted as pairing a phone with a speaker. A word that was accepted among all participants was *sync*.

When discussing the proposed symbols, participants agreed that the symbol should be simple and easily referable using a single word. Several symbols proposals were discussed, including a modified version of the existing M-button, as well as symbols illustrating a house, a chain and the letter C representing *connect*.

# 8. Concepts

Based on insights from the ideation phase three main concepts were developed speaking to different needs and functions. A common factor in all concepts is Bluetooth as a connectivity method and the integration of Auracast technology. This allows customers to only buy speakers when wishing to expand the music experience in their environment, thus removing the Heddon as a complementary product that does not play music. These concepts are positioned to fit in between Marshall's current portfolio of portable speakers, less rugged than Emberton and Middleton, but providing greater portability than Kilburn.

## 8.1. Brixton

Brixton was developed as a concept aimed at expanding the sound experience while ensuring a simple installation and charging process. The design is inspired by Marshall amplifiers, which are closely associated with the brand and widely recognised as statement pieces. These amplifiers are typically large and often stacked to intensify the sound output. Brixton adopts the visual of amplifiers while reducing the overall size, making the product more accessible to a broader customer base while preserving its distinctive character.

Brixton consists of two speakers stacked on top of each other, where the sound is enhanced when both units are combined. The bottom speaker functions as a stationary unit, while the upper speaker is designed to be movable. The two speakers are synchronised using Auracast and are pre-configured to each other, allowing the user to complete the installation process only once. A concept sketch is presented in Figure 32.




*Figure 32: Concept sketch of Brixton*

The bottom contains an Auracast transmitter and is constantly connected to a power source. The upper speaker contains an Auracast receiver, allowing it to receive a broadcast signal from the

bottom unit. Additionally, other devices with Auracast receivers can be synchronised with the bottom speaker. Both speakers are equipped with Bluetooth, enabling them to function separately if needed. While the top unit is provided with a USB-C charging port, it can also be charged directly through the bottom speaker when docked. Charging is facilitated through contact points located in the feet of the upper speaker and connectors on the bottom unit.

The upper speaker is designed to signal movability through a hard handle with wooden detailing, reinforcing the retro aesthetic associated with amplifiers. It is intended to be movable rather than fully portable, thus water resistance and not water proof.

## 8.2. Bexley

Bexley was developed to simplify the installation process, while enabling flexible charging. Among the three concepts, Bexley draws the strongest inspiration from Marshall's existing product portfolio by keeping recognisable design elements associated with the brand identity. However, the overall form factor differs by adopting a more quadratic geometry rather than the traditional rectangular shape. This design choice aims to enhance stability while maintaining a compact size, making the speaker suitable for being moved around. And also helps differentiate it from current speakers, since it comes with new functions it should also be clear that this is not the same speaker as the ones before.

Bexley consists of two main components, a speaker and an accompanying docking station. The docking station is designed as a stationary element and serves a dual purpose. First, it functions as a charging station for the speaker. Second, it acts as a central hub that facilitates synchronisation between multiple speakers using Auracast. As the docking station is intended to remain visible within the room, emphasis was placed on designing it as a standalone statement piece. Ensuring visual appeal even when the speaker is not docked. Concept sketches are visualised in Figure 33.

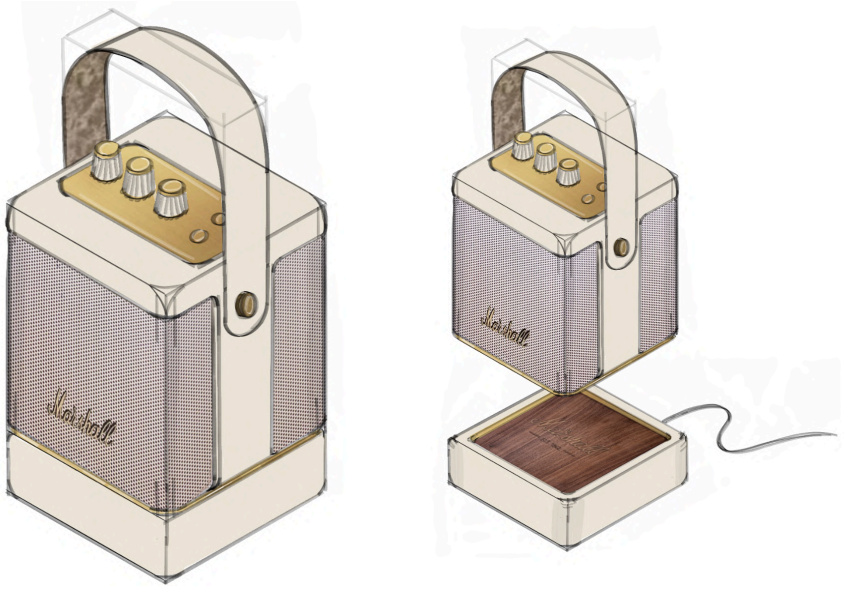


*Figure 33: Concept sketches of Bexley*

The speaker is equipped with Bluetooth and an Auracast receiver, while the docking station integrates Bluetooth and an Auracast transmitter. Similar to the Brixton concept, the two units are delivered pre-configured to reduce set up complexity. Audio playback is initiated via Bluetooth through the docking station, which subsequently broadcasts the audio signal to all connected devices using Auracast. This enables the speaker to be freely moved around within the effective range of the broadcast signal while maintaining synchronised playback. But music playback can also be initiated directly to the speaker itself via regular Bluetooth set up.

The speaker supports both wired charging through the USB-C port and contact based charging when placed on the docking station. The movable part is provided with a flexible handle, featuring crushed velvet on the inside, inspired by existing handles on their portable speakers. Furthermore, the speaker is designed to be water resistant rather than fully waterproof, reflecting its intended use as a movable device rather than a fully portable, rugged product.

## 8.3. Sutton

The Sutton concept differentiates the most from Marshall's design language, between the three concepts. While it reserves some retro elements, such as Marshall's iconic control knobs, it introduces a new form factor. This redesign aims to clearly differentiate the product from existing speakers. The redesign is intended to create a clear differentiation from existing speakers, thereby reducing the risk of user confusion arising from similar appearance across products with different functions.

As with the previous concepts, the Sutton speaker incorporates charging via a dedicated charging station. This solution was developed to address common user frustration associated with charging. A sketch of the concept is presented in Figure 34. Within this concept, variations in knob placement were also explored as part of the design development process.




*Figure 34: Concept sketches of Sutton*

The charging station is intentionally minimalistic, allowing it to be easily removed or stored when not in use, although doing so limits its functional relevance. Both the Auracast transmitter and receiver are integrated into the speaker. Playback is initiated via Bluetooth, and other speakers equipped with Auracast receivers can connect seamlessly to the Sutton unit.

Furthermore, the speaker includes a removable handle, inspired by the design of Marshall's Middleton speaker. The handle is constructed from leather, with a crushed velvet lining, that can be easily attached or detached depending on the user's preference for portability.

# 9. Development

The concept selected for further development was Bexley, as it aligned well with Marshall's existing product portfolio while addressing several identified customer needs. The decision was further supported through discussion with the design team. Brixton was considered too costly to develop, while Sutton was assessed to deviate too far from Marshall´s established design identity.

Particular focus was placed on the docking station accompanying the Bexley speaker, as it showed significant potential for both forward and backward compatibility. This aspect was considered as highly important, as it allows existing Marshall customers to get continued value in their current products, rather than replacing them due to incompatibility. From a sustainability perspective, this supports product longevity and reduces unnecessary product replacement.

Following the concept selection and initial application layout drafts, the concept was further refined with focus on charging, synchronisation and ecosystem integration.

## 9.1. Concept Development

A major part of the concept development focused on transforming the docking station from solely a charging accessory into a central hub within the ecosystem. Early discussions concerned where the Auracast transmitter should be placed to bring most value and least friction for the customer. After discussions and identifying different scenarios the transmitter was decided to be integrated to the docking station itself. The docking station remains continuously connected to power, making it accessible, visible and a central product in the home environment. This also supports the intention of designing the docking station as a statement piece that follows Marshall's established design language, rather than as a technical accessory intended to be stored away. By placing the transmitter in the docking station enables a more stable and stationary broadcasting point whiting the ecosystem. Since the docking station remains centrally positioned in the home, the connected speaker can move independently without interrupting the broadcast. This allows the movable Bexley speaker to leave the home and operate separately via Bluetooth while the docking station continues broadcasting audio to remaining connected devices.

Another key development area concerned simplifying the synchronisation process between speakers and the encrypted Auracast broadcast. The existing solution relies on manual button-based pairing between devices, guided by the accompanied application. One device opens

the broadcast and additional speakers manually join it though button presses. The proposed concept instead explored whether synchronisation could be facilitated directly through the docking station by the use of Near Field Communication (NFC) technology. NFC enables short-range tap-to-connect interactions between nearby devices (Kjell & Company, 2019). By placing speakers onto the docking station, synchronisation could potentially occur automatically. Thus reducing friction while still maintaining a physical and tactile interaction with the products.

Charging functionality was explored extensively, looking at wireless solutions compared to contact-based alternatives. Wireless charging was initially considered but excluded due to lower charging efficiency, heat generation and strict alignment requirements (Endesa, 2025). These limitations were considered particularly problematic for a universal charging platform intended to support multiple speaker models.

Instead, pogo pins were selected as the preferred charging solution. Compared to wireless charging, pogo pins provide reliable alignment and more efficient charging while still enabling clean and simplified docking interaction. The long-term intention was that the integration of pogo pins could be compulsory for future generations of Marshalls portable speakers. Thus enabling a unified charging ecosystem where multiple devices can be charged though the same docking station. Despite being contact based, waterproof variants are commercially available and therefore making them suitable for the environmental requirements of the concept.

To support changing functionality and create a more integrated product experience, a power supply unit (PSU) was integrated into the product itself. Since a PSU is required for the charging system and the docking station is intended to support multiple speaker models, this was considered to be the most suitable solution. Similarly as with the Heddon today. This also allows the cable design to align with the overall colour scheme and visual identity of the product, reducing the need for customers to purchase external accessories separately.

Several concepts for backwards compatibility with older speaker models were also investigated. One proposed solution involved integration of a USB-C connector into a wall of the docking station, allowing speakers to slide into position and connect. However, differences in charging port placement between models made it difficult to create a universal solution. Alternative concepts involving stepped platform levels were also explored, but were ultimately considered too complex and infiltrated too heavily on the design. Separate modular accessories were considered where interchangeable interface pieces could be inserted to adapt the speaker to align with the docking station. This solution however, introduced a large number of small accessory components, making it less practical from both a usability and product management perspective.

Another idea involved integrating a retractable charging cable directly into the docking station, allowing manual connection when needed. Although this solution required a more active user interaction, it was considered one of the most practical approaches for enabling backwards compatibility. Multiple iterations and sketches were developed throughout this exploration process.

To further support backwards compatibility with older Marshall and third-party products, RCA connectors were considered as an interface for legacy audio devices. RCA remains a widely used standard for connecting audio equipment, such as vinyl or other analog systems that Marshall aims to support.

## 9.2. Control Panel Layout

The early concept sketches of the speaker did not include detailed visuals of the control panel. Instead, placeholders were used to indicate its intended location. Iteration of the control panel layout was therefore conducted after the concept selection phase. This process began with examining control panels from Marshall's existing product range. Both to determine essential functions and take inspiration from their design language. Pictures of control panels are collected in Figure 35.




*Figure 35: Control panels of several speakers from different generations*

A self evident decision was to include a Bluetooth button with the classic Bluetooth symbol, which enables connection between speaker and a mobile device. This choice was straightforward, as the symbol is widely recognised and contributes to user familiarity, thereby enhancing intuitive interaction.

Subsequently, different configurations for the power button were tried out. Two main alternatives were considered, a push button or a power switch. If a switch were to be used, it would need to bounce back to its original position rather than remaining as fixed on either on or off position. Without the bouncing back function, it could create issues when the portable unit goes into power saving mode, as the switch might remain in the "on" position while the device has actually powered down. Ultimately, a push button was selected in order to avoid confusion with stationary units, where switches remain in fixed position to indicate power status.

Iterations for volume adjustment were also explored. However, the decision clearly pointed towards using Marshall's characteristic knobs. These are already implemented in some of their portable speakers, ensuring their suitability for a portable product. Additionally, the knobs contribute to maintaining brand identity and were positively commented by all observation participants.

Lastly, the design of a symbol representing the synchronisation function was explored. Several alternatives were explored, including adaptations of Marshall's existing "M-button", used to preset functions. This could be done by adding a synchronization label or modifying the symbol into an "M-sync" symbol incorporating wave elements. These options were rejected, as they were considered likely to create confusion with one button addressing multiple functions and too similar symbols. Other symbols discussed during the workshop, such as "link" and "connect" were also evaluated. The final choice was a plus (+). This symbol was considered initiative, as it conveys the action of adding another device, and it can be described with using a single term. Furthermore, the same symbol is used within the application, creating consistency between physical and digital interaction.

Initially, the use of a plus symbol was avoided due to its similarity to the symbol used increasing volume. However, as the concept evolved, the synchronization button was relocated to the docking station. Since the docking station does not produce any sound, the risk of confusion with volume adjustment was eliminated. Sketches of design alternatives for each function are presented in Figure 36.

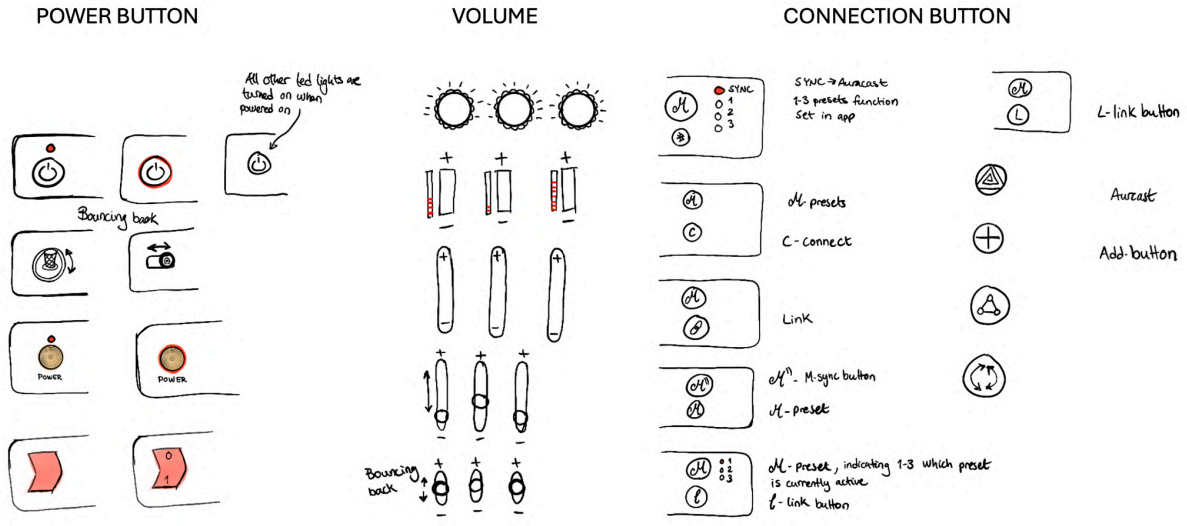



*Figure 36: Sketches of control panel layout*

## 9.3. Application Layout

Based on proposed layouts and insights gathered during the workshop discussion, an initial version of the application layout was developed. The complete overview of all pages are presented in Appendix E. The first version was presented to an employee working with user interface and user experience at Marshall, whose feedback was used in further refinements.

The first page developed was the homescreen. It became evident that all devices connected to the user's phone should be visualised on this screen and made easily accessible. In addition, the connection status to Heddon is displayed, given that Heddon is active within the system. Users are given the option to personalise device names while still having access to their original product names.

Certain elements were identified as essential across all pages. These included a home icon and a menu icon. Furthermore, a master volume control is available when multiple speakers are connected within the same system, enabling users to adjust the overall sound.

On the individual device page, the connection to Heddon is clearly stated. This page also provides guidance on how to use the device and its various functions. In addition, standardised product information, such as product identification number and firmware updates, is available.

The last page was the menu page, which the current application does not have. This page integrates categories that have previously been distributed across different sections of the application, but with a menu the accessibility and navigation is now improved. Moreover, the design of the menu page strongly reflects Marshall's brand identity.

The feedback provided by the UI and UX designer was incorporated into the final layout proposal. Recommendations regarding typography, sizing and spacing within the interface were addressed during the refinement process. Additionally, the integration of Marshall's visual identity within the application was highlighted as a positive and valuable aspect of the design.

# 10. Final Concept

In the following chapter the final speaker concept Bexley is presented as well as a finalised proposal of application layout and sustainability aspect.

## 10.1. Speaker

Bexley is a two-component unit consisting of a speaker and an accompanying docking station, which functions both as a charging base and as a central hub for connecting additional speakers. An overview of the unit is presented in Figure 37.




*Figure 37: Bexley in cream and black with accompanying docking station*

The design of the speaker draws heavily on Marshalls brand identity as well as on both current and legacy products. In particular, it incorporates design cues from the Stockwell, Heston and Heddon models, alongside elements inspired by the earlier Uxbridge speaker. This synthesis of references results in a product that balances familiarity with new features.

The combined influences shape Bexley into a stable yet portable speaker, characterised by a robust form that conveys durability while supporting ease of movement within the home environment. At the same time, the integration of additional functional features enhances the overall user value, positioning the product as both practical and aligned with the brand's design language.

The exterior of Bexley is constructed from leather, applied both at the top and side surfaces. This creates a cohesive visual continuity with the handle, which is likewise leather wrapped. The handle interior has a velvet finish, which makes it a softer contrast and aligns with existing products. The speaker units are protected by a contemporary mesh featuring a pattern of rounded perforations, resulting in a smooth and visual streamline surface.

The docking station is also constructed of leather, however when the speaker is removed a contrasting material composition is revealed. This includes a brass surface combined with dark

walnut wood, into which the Marshall logo is engraved. This material juxtaposition is intended to transform the docking station into a distinct visual centerpiece, while simultaneously maintaining a retro-inspired aesthetic within a modern context.

At the base of the speaker, a recessed indention facilitates precise and intuitive placement onto the docking station. The brass detailed at the bottom aligns with the corresponding elements on the dock, contributing to functional stability. This brass finish is consistently applied across components, including the handle fastenings and the interface of the control panel. The control panel itself is primarily constructed from brass, complemented by selected silicone elements that engage both visual contrast and tactile interaction. The speaker is available in black and cream colorways, aligning with Marshall's established products palette. The introduction of wood as a material element represents a subtle expansion of the brand's material language, while remaining consistent with its overall identity and design heritage.

The Bexley speaker has overall dimensions of 180 x 120 x 120 mm, see Figure 38 for more detailed dimensions. It is designed to meet the IPX4 standard for water resistance, providing protection against splashing water from any direction. This level of ingress protection supports its intended use within both indoor and transitional outdoor environments. Consequently, the speaker can be moved freely throughout the home, including placement in semi-exposed areas such as balconies, without concern for light rain or incidental moisture exposure.




*Figure 38: Bexley with dimensions of different parts*

The docking station incorporates an Auracast transmitter broadcasting audio to the speaker, which is equipped with a corresponding Auracast receiver. In addition, both the docking station and the speaker support Bluetooth connectivity, enabling compatibility with a wide range of devices and supporting a positive user perception. Charging is facilitated through pogo pin connectors located at the base of the docking station, see Figure 39. This approach allows for more efficient and reliable energy transfer compared to fully wireless charging solutions, while maintaining ease of use. The speaker also includes a USB-C port, ensuring compliance with contemporary charging standards and providing alternative charging methods if needed. Furthermore, this feature allows the speaker to be used independently of the docking station, thereby supporting portability and enabling use outside of the docking station area, such as in a vacation cabin.

Moreover, the docking station integrates NFC technology to support simplified device pairing. By placing a speaker equipped with NFC integration on top of the docking station, the device automatically connects to the broadcast. This introduces an intuitive tap-to-connect interaction for users of Marshall products. While current Auracast-enabled Marshall speakers may not yet utilise NFC, this feature has been included to ensure future compatibility. For synchronisation with existing Marshall speakers that support Auracast but lack NFC, a manual pairing process is provided. This involves activating an "Add" function via a button, with a "+" symbol, located on the side of the docking station, which opens an encrypted broadcast channel visually seen by a pulsating light, the same light seen on the Heddon today. Additional devices can then connect by pressing the designated M-button on the speaker intended for pairing.

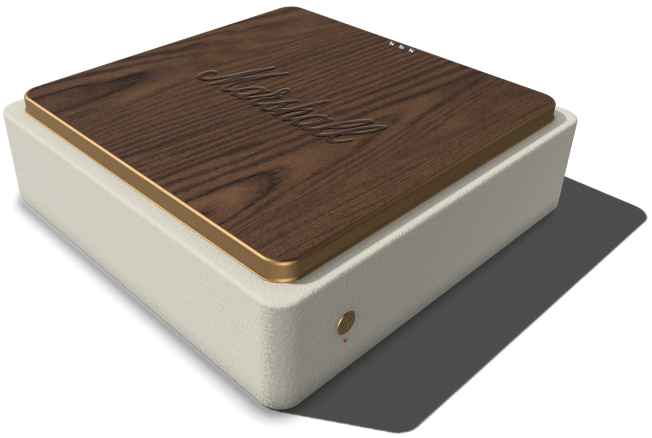


*Figure 39: Docking station with add button on the right side and pogo pins at the back*

The docking station incorporates two USB-C outlets to ensure backward compatibility with existing speaker models, as well as enabling the charging of mobile devices, addressing the relatively high energy consumption associated with Bluetooth usage, see Figure 40. The docking station can therefore also function as a centralised charging hub within the home environment, providing a dedicated and consistently accessible location for charging. The PSU is integrated within the docking station and is therefore accompanied by a consistently designed cable. Additionally, RCA input and output connections are included to further extend compatibility with legacy audio systems.



*Figure 40: Docking station from the back showing AC in, RCA and USB-C ports*

The control panel includes rotary knobs for volume, bass and treble adjustments, each surrounded by LED indicators that display the corresponding level, see Figure 41. The speaker is powered on via a dedicated power button, adjacent to which a visual indicator bar displays the battery status. When charging, the battery indicator displays an animated charging sequence, in which illuminated segments progressively pulsate to indicate increasing charging levels. Once a level is fully charged, it becomes steadily lit, while the subsequent level begins to pulsate, providing a clear visual representation of the charging progression.




*Figure 41: Control panel layout*

Media playback is controlled through a multifunctional knob that supports play, pause and track navigation. This knob exists on multiple devices today and has clear haptic feedback when interactive with. Pressing the control initiates or pauses playback, while lateral movement allows the user to skip forward or return to the previous track.

A dedicated Bluetooth button, marked with the Bluetooth symbol and paired with an LED indicator, communicates the device's Bluetooth connectivity status. Additionally, the “M”-button is illuminated when the speaker is connected to the docking station and is able to receive an audio broadcast via Auracast.

## 10.2. Application layout

The redesigned application layout centered around a primary home screen that integrates the key functionalities while simultaneously presenting an overview of the current device configuration. The interface has been simplified by removing the earlier bottom navigation bar, replacing it with

a home button and a menu icon positioned in the upper corners of the screen. This design decision increases the available display area while ensuring that essential control remains consistently accessible throughout the user experience. Furthermore, a plus sign is displayed on the right side, allowing users to connect new devices. This plus sign is repeated from the docking station to create consistency and familiarity between the physical product and the application.

The first home screen layout, illustrated on the left in Figure 42, presents all units installed and paired with the current device. The Woburn III is prominently positioned at the top and displayed at a larger scale, accompanied by a volume bar to indicate that it is the actively playing speaker. Other speakers are represented at a smaller scale, and portable speakers are also dimmed down when powered off signaling their availability but inactive state. Each device is labeled with both its product name and a user-customisable, allowing for contextual identification based on usage or location. These labels persist across the application and can be modified directly through simple text input. The system is similarly editable, ensuring constancy and flexibility in naming conventions. The second screen, in Figure 42 illustrates the screen after installation of Bexley, when only the Bexley speaker is connected to Bexley Hub. while still displaying your previously installed speakers below.




*Figure 42: Home screen showing either when having separate units or a system playing together*

The two rightmost screens in Figure 42 illustrate a scenario in which a system configuration is actively playing music via Auracast. In this case, the Bexley Hub functions as the transmitter, broadcasting audio to connected speakers. The Kilburn remains intentionally disconnected, which is visually communicated through an on/off toggle positioned to the left of the individual unit. This interaction allows users to easily include or exclude speakers from the system,

introducing a tactile and intuitive control mechanism that aligns with Marshall's emphasis on physical interaction. It also supports use cases where users may wish to play different audio across separate devices.

Each individual speaker within the system is equipped with a dedicated volume control, enabling precise adjustments directly from the interface. The volume bar transitions visually from white to a brass finish, and is adjusted via a circular control element. For portable speakers connected to the system, battery status is displayed in percentage next to a battery icon. The indicator switches from grey to yellow when the charge level drops to approximately 20% and further to red when hitting 10%, providing clear feedback on device status.

At the bottom of the interface, a persistent media control bar displays the current playing content along with a master volume control that adjusts all connected speakers simultaneously. Expanding this bar reveals three configurable master presets, allowing users to save and quickly apply combinations of bass and treble across the system. This feature enhances usability by supporting personalised audio configurations tailored to different listening contexts.

Furthermore, Figure 43 illustrates the layout addressing the three cases discussed during the workshop. The leftmost example presents a scenario in which a speaker is disconnected from the hub. In this state, both the Actons image and label is visually dimmed, and an exclamation mark replaces the standard connection status icon to indicate an issue. The user can click on the device by the arrow, image or label, to navigate to its dedicated page, where more detailed information regarding the connection problem is available.




*Figure 43: Home screen showing the three cases, a speaker has lost connection, a portable speaker is out of battery and lastly the main hub has lost the connection*

The second example in Figure 43 depicts a portable speaker that has run out of battery. Similar to the previous case, the device is dimmed to signal its inactive states. However, the connection status icon is replaced with a battery indicator. In both of these cases, despite the occurrence of errors, the devices remain visible within the system overview. This design choice reflects the user's intended system configurations, signalling that the device belongs to the system but is temporarily unavailable due to external constraints.

Finally, Figure 43 illustrates a situation in which the Bexley Hub has lost connection. As a result, the visual linking lines between devices disappear and the connection status switches to "off", indicating that the system is no longer actively connected. Although all units remain visible on the screen, only the Woburn speaker continues playback, automatically switching to a Bluetooth connection. Consistent with the primary home screen design, the Woburn is displayed at an enlarged scale to emphasise that it is the current active device.

When clicking on either the image, device name or the arrow positioned to the right of the name, the user is directed to an individual device page, see Figure 44. Figure 44 presents both the individual page of the Bexley speaker, as well as the layout of the individual page of the Bexley Hub. Upon entering this view, the home icon in the upper left corner changes from white to brass, indicating that the user is no longer on the home screen.

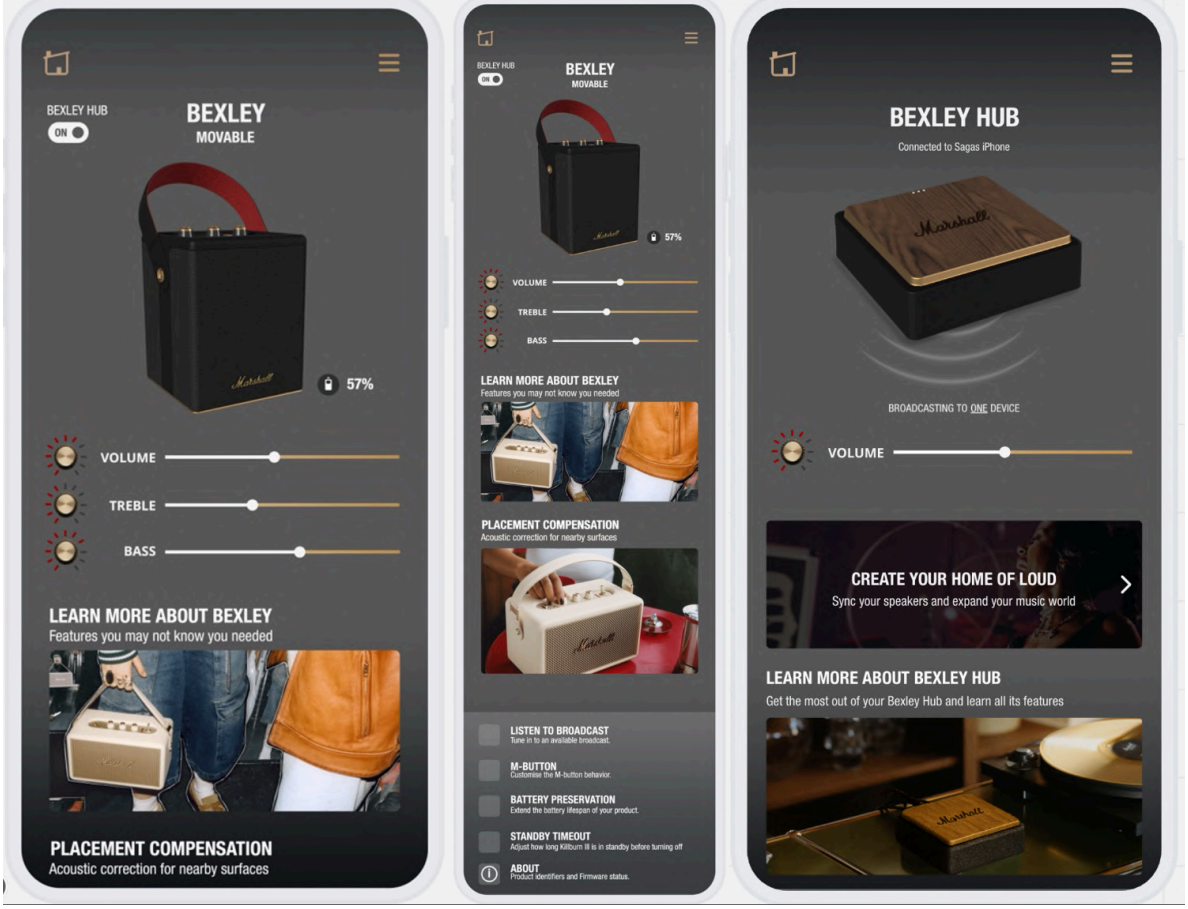



*Figure 44: Screen for one separate device*

This page builds upon the existing device view found in the current Marshall application but extends its functionality by incorporating system-level context. The device's connection to the system is clearly displayed though a connection status symbol and the system name, both positioned to the left side. Users can easily disconnect the device from the system by toggling this status off. On the Bexley Hub page, instead, informs the user how many speakers are connected to the Hub as well as to what mobile device it is connected to. Within this page, users can manipulate the speakers volume, bass and treble through interactive sliders, similarly designed as on the home screen. Each control is complemented by a corresponding knob positioned on the left side, which moves synchrony with the slider input. This design decision reinforces the brand identity by translating tactile qualities of Marshall's physical products into the digital interface. While the Hub page presents the option to control the master volume. Further down the page, users can access additional information about the speaker This content was previously presented immediately after the installation of a unit. However, it has been relocated to this section to improve the overall user experience. By postponing informational content, the design prioritises immediate access to music playback after a successful installation, thereby reducing interruptions and enhancing perceived value.

When clicking on the menu icon located in the upper right corner, a panel slides in from the right side of the screen, revealing the application menu, see Figure 45. This menu introduces a revised visual design that closely aligns with the Marshall brand identity. The interface emphasises strong visual elements, where primarily navigation categories are accompanied by bold, expressive imagery to create a more distinctive and brand-consistent experience.




*Figure 45: Menu screen moving in from the right side and digital records feature*

A new feature, *Digital Records,* is also introduced within this menu. This feature is created to engage the users with the people behind the products. Through *Digital Records*, Marshalls employees can share their most-played or personally curated tracks over a given time period. An integrated AI tool then compiles these selections into a curated record selection, making them accessible to users.

This feature not only enhances users' engagement but also reflects a core aspect of the Marshall brand, its commitment to support and promote emerging artists and creators. By embedding this feature within the application, the brand is able to extend this value proposition into the digital domain. This offers users a dynamic and interactive way to discover new music while fostering a sense of community and authenticity.

Figure 46, illustrates the final step in the step-by-step installation guide. While the preceding steps remain consistent with the current procedure, several improvements are recommended, including enhanced contrast of loading elements, clearer indication of current status and more informative content.

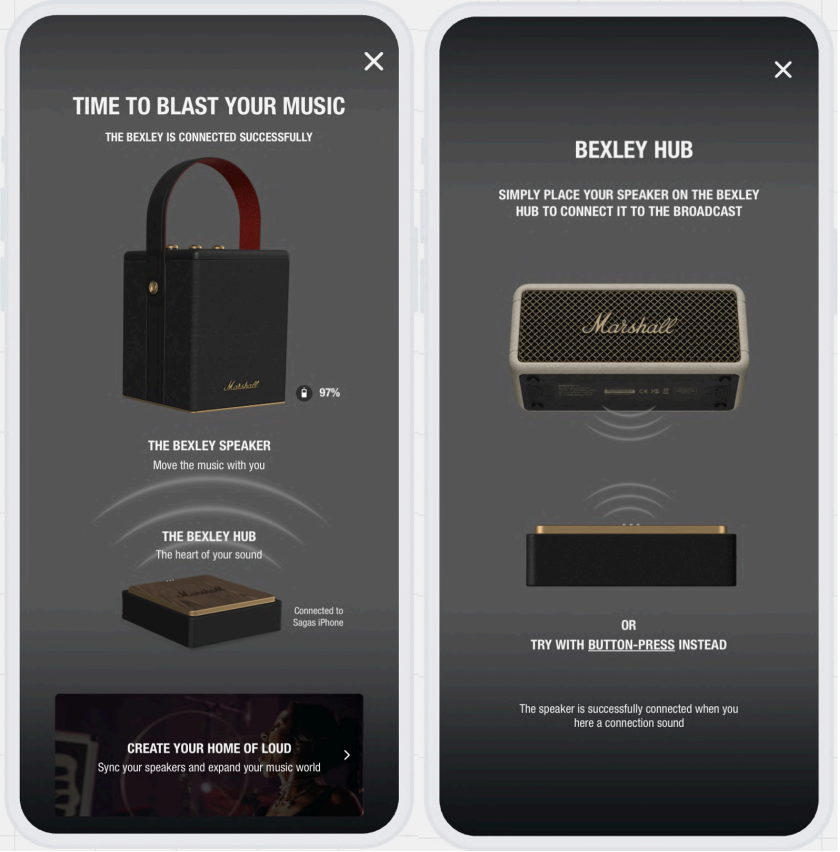



*Figure 46: Last step in the installation process, if wished to add more speakers to the system*

In the first picture in Figure 46, the user is informed about how the setup functions through illustrations that demonstrate both the mobility of the speaker and the role of the Bexley Hub. The hub serves as the intermediary that communicates with the user's phone, indicated by the text "*The heart of you sound*" and the informative text "*Connected to Saga's iPhone*". These elements serve to clarify both which device is responsible for communication and the specific component

in the setup to which the phone is connected. The speaker receives the broadcast supplied by the hub, allowing the user to move the speaker wherever is needed, indicated by the text "*Move the music with you*". Further, at the final stage the user is presented with two possible pathways. The first option is to exit the process by selecting the close icon in the upper-right corner, thereby proceeding directly to music playback. The second option is to select the "Create your home of loud" button located at the bottom of the screen.

This option initiates the Bexley Hub connection process and enables adding further speakers to the system, as seen in picture two in Figure 46. The interface provides the possibility to add a device via NFC, provided that the speaker supports this technology. NFC compatibility is clearly indicated during the installation process and on the product packaging. Synchronisation takes place by placing the speaker on the Bexley Hub. The installation is successful if there is a confirmation sound. NFC compatibility is clearly indicated during the installation process and on the product packaging.

If the device does not support NFC pairing, synchronisation must instead be performed through a manual button-press method. To receive guidance for this procedure, the user can select the underlined term *button-press* on the right picture in Figure 46, which redirects them to further instructions.

If the speaker does not have NFC, the user is directed to an alternative synchronising process, consisting of three steps, as illustrated in Figure 47. Syncoronisation is initiated by pressing the plus button on the hub, as shown in the first screen in Figure 47. Subsequently, the user presses the M-button on the speaker, as illustrated in the middle screen. Finally, the success of the synchronization is confirmed when the M-button is illuminated, as shown in the last screen.




*Figure 47: Synchronising process with a speaker of an older version*

A process bar is displayed at the bottom, indicating the current step in the process as well as the number of steps remaining. Throughout the process, troubleshooting support is continuously available, enabling customers to access assistance and further information. This ensures that the process can be completed with minimal friction.

## 10.3. Sustainability

Marshall works strongly towards being sustainable, by prioritising responsible material selection and manufacturing processes. Bexley will adopt these established practices to align with the company's environmental standards and ensure compliance with Forest Steward Council (FSC ) certification requirements. In line with Marshall, Bexley will incorporate materials such as recycled plastics, recycled metals, FSC certified wood, water-based PU leather and PFAS-free components. These choices contribute to reducing reliance on virgin resources, lowering associated environmental impact and decreasing carbon emission compared to conventional material production.

Further, a goal of Bexley is to ensure both forward and backward compatibility, as previously discussed. This approach allows the product to function seamlessly with both newer and older devices, thereby reducing the need for replacement due solely to compatibility limitations. By avoiding premature product replacement, it contributes to extending the product's lifespan and minimising electronic waste.

# 11. Discussion

In the following chapter the final concept is discussed, sources of error and future work.

## 11.1. Final Concept Discussion

One of the most challenging decisions concerned whether the docking station should incorporate Bluetooth or Wi-Fi as a connectivity method. Ultimately, Bluetooth was selected, which meant that expressed user frustration regarding sound interruption were not addressed. However, another frustration that was brought up was difficulties hearing when answering calls while music played in the background. By having Bluetooth, these frustrations are solved since the music pauses when answering the call. This decision was primarily based upon user's perceptions of Bluetooth as a straightforward and intuitive technology. Additionally, the speaker was conceptualised as a portable device, which is strongly associated with Bluetooth, and it was assumed that introducing multiple connection methods within the same product could lead to confusion. Finally, Bluetooth was considered due to its reach of a broader customer base.

Another challenging decision consideration involved whether to incorporate the Auracast symbol on a dedicated button to indicate when the speaker was listening to an Auracast broadcast. In the final concept, this function was instead integrated into the existing M-button. The decision not to implement the Auracast symbol was based on users' hesitancy towards unfamiliar technologies. Rather than emphasising the underlying technology, presenting the function itself was identified as a more effective approach for supporting user understanding and acceptance.

## 11.2. Sources of Error

One source of error was identified during the competitive analysis. In the initial stage, user reviews were examined to understand general perceptions of competing brands and products. However, it was noted that users who posted reviews were either very positive, leaving a five star review, or very negative, leaving a one star review. This pattern reflects a typical human behaviour, where users with extreme opinions are more motivated to leave a review, while users with neutral opinions typically remain silent. As a result, the reviewed data may not accurately represent the average user experience, but rather the extremes of the spectrum. Users with more neutral views are more effectively captured through methods such as interviews or surveys, where participation does not depend on voluntary motivation.

The observation scenarios in this study may be considered somewhat unrealistic. Purchasing and installing five new speakers simultaneously is not common, which may have contributed to increased levels of frustrations compared to a normal installation process. Further, the observations were conducted early in the year in Stockholm, where weather conditions are generally unfavourable to spend time outside. Consequently, limited information about use case III was a result, as the speakers were not relocated outdoors.

Furthermore, it was noted that the feedback collected in the domestic setting was more substantial than that obtained in the office environment. One possible explanation for this

difference lies in the data collection methods employed. Participants in the domestic setting received individual letters, whereas participants in the office setting were provided with a shared space for leaving feedback. This represents a typical case in which shared responsibility results in no one's responsibility. If the observation were to be conducted again, the data collection method would have been reconsidered to ensure higher volume of feedback. Implementing recurring digital surveys, for example, on a weekly basis could have been a more effective approach.

Lastly, the influence of social desirability may have occurred during interactions with participants in both the user-study and workshop. Participants may not have expressed their genuine opinions, instead providing responses perceived as desirable. Consequently, the answers received may reflect what participants believed was expected rather than their true perspectives.

### 11.3. Future Work

A part of future work would be to conduct an additional iteration round involving users, trying out trigger material, to further evaluate the proposed concept. The use of trigger material would help with assessment of design decisions by anchoring them to user feedback. For instance, such testing could have examined whether the plus button on the docking station is perceived as intuitive. Trigger material would have enabled a more thorough validation and refinement of the concept, strengthening the design choices. However, due to time constraints, such testing was not conducted, and earlier user interactions were therefore used as the foundation for design decisions.

A decision was made not to develop a solution to when several people want to stream music from the same device, since Spotify Jam worked well. In Sweden, Spotify is the most widely used music-streaming service (Internetstiftelsen, 2019). Further, Spotify is the dominant music streaming service in Europe, with a market share of 56% (Apple, 2024). Hence recruiting individuals who used other streaming services proved difficult, resulting in all participants in the observations and interviews being Spotify users. A part of future work could be to examine users of other music streaming platforms to gain their perspectives. This may have led to a result that development of a solution for music sharing in fact was necessary.

## 12. Conclusion

To conclude how Marshall can integrate new technologies seamlessly into the speaker experience, based on the conducted user study, the solution requires both a thoughtfully designed speaker and a complementary intuitive application interface. Bluetooth was selected as the connectivity technology to compensate for the absence of an Auracast interface, with the aim of reaching a broader customer base. The speaker should incorporate clear signifiers that communicate its intended features, focusing on the function it enables rather than the underlying technologies. Meanwhile, the application, as indicated by the user study, should guide and enhance the users understanding of the ecosystem. Furthermore, it should minimise dependence on purely digital interactions by maintaining a balanced integration of physical and digital

elements. Thus aligning the tactile qualities characteristics of Marshall's existing portfolio and the increasing technical complexities driven by emerging technologies.

The proposed solution meets the three user cases as outlined below.

I. Situations where multiple speakers are intended to sync and play simultaneously

The Bexley concept enables users to create a centralised multi-speaker system. Multiple speakers can synchronise and play simultaneously though an Auracast broadcast transmitted from the centerpiece the Bexley Hub. Future speakers can join the broadcast through a simplified tap-to-connect interaction enabled by NFC. Earlier speaker generations can connect though a manual button-based pairing process, opening the broadcast for additional speakers. Furthermore, RCA outputs allow legacy products to be integrated into the system as well.

The Auracast technology enabling Bexleys features and system set-up is further visualised and supported though an accompanying application designed to align with Marshall´s brand identity. The application aims to increase user understanding of the ecosystem and provide added value throughout the usage. It is designed to function as a complementary tool rather than a required part of the user experience.

II. Situations where multiple users wish to play music together on the same speaker(s)

The proposed solution primarily relies on Spotify Jam as the established method for collaborative music interaction. This reduces the need for Marshall to develop an entirely separate solution, especially within the European market where Spotify holds a strong market presence.

III. Situations where speakers are used across different locations, both in and around the home and potentially outside the home

The Bexley speaker is designed to support mobility through a stable base form, an integrated handle and accessible charging via its dedicated docking station. The speaker automatically connects to the Bexley Hub and tunes in to the Auracast broadcast. Thus enabling seamless movement throughout the home while maintaining continuous audio playback. This allows the music experience to follow the user across different environments, enlarging the music world. The speaker is designed to function not only as a movable speaker but also a statement piece that aligns with Marshall's established design identity and visual presence.

In conclusion, this study demonstrates that integrating emerging connectivity technologies into speaker systems requires more than simply introducing new technology. It demands careful consideration of the user and a balance between technological capabilities and how they are presented to ensure understanding. The findings showed that users prioritise simplicity, familiarity and intuitive interaction. Ultimately, this work emphasises that successful innovation lies not only in advanced technology, but in how seamlessly it can be understood, adopted and valued by users.

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# Appendix A: Observation Material

Before the observations were conducted, events that may occur during the observation were noted in Table 1. Next to the events, a column counting how many times every event occurred and another column with additional comments were written down. Along with table 1, a few questions were added at the end of the observation to conclude and summarise feelings and experiences. All observation materials were designed in Swedish, since all participants were fluent in Swedish.

*Table 1: Events that may occur*

| **Vad händer** | | **Antal gånger** | **Kommentarer** |
|---|---|---|---|
| Trycker på fel knapp/element flera ggr | | | |
| Letar efter funktioner/knappar | | | |
| Behöver försöka flera gånger för att lyckas | | | |
| Undviker funktioner som känns osäkra | | | |
| Tvekar | | | |
| Suckar | | | |
| Glädjeyttringar | | | |
| Frustration | | | |
| Irritation | | | |
| Kroppsspråk | Glädje | | |
| | Trötthet | | |
| Känsla av haptisk respons | Negativ | | |
| | Positiv | | |
| Känsla av auditiv respons | Negativ | | |
| | Positiv | | |
| Känsla av visual respons | Negativ | | |
| | Positiv | | |

Tid för installation:

FRÅGOR TILL EFTER
Hur lätt kändes installationen? 1-5

Var ni medvetna om att det fanns olika uppkopplingsalternativ?

Känner du att du skulle kunna förklara det här för någon annan?

Var det någon funktion tyckte var extra hjälpsam? (Haptisk, visuell, audio respons)?

Var det något steg du nu i efterhand inte förstod varför du gjorde?

Hur tyckte du balansen mellan app-interaktion och manuell interaktion var?

Vad var lättast?

Vad var det mest frustrerande?

Vilken typ av feedback gav mest tillit?

Var det någon typ av feedback som kändes störande?

Fanns det information som saknades?

Ändrades din känsla under installationens gång? isåfall hur?

---

While the observation took place, information about the participants' user experience was gathered through interactive material. At the homesetting, each participant received six letters that they got to answer evenly distributed during the test month. Layouts of the six letters are presented in Figure 1 and Figure 2.




*Figure 1: Three first personalised letters for participant one*

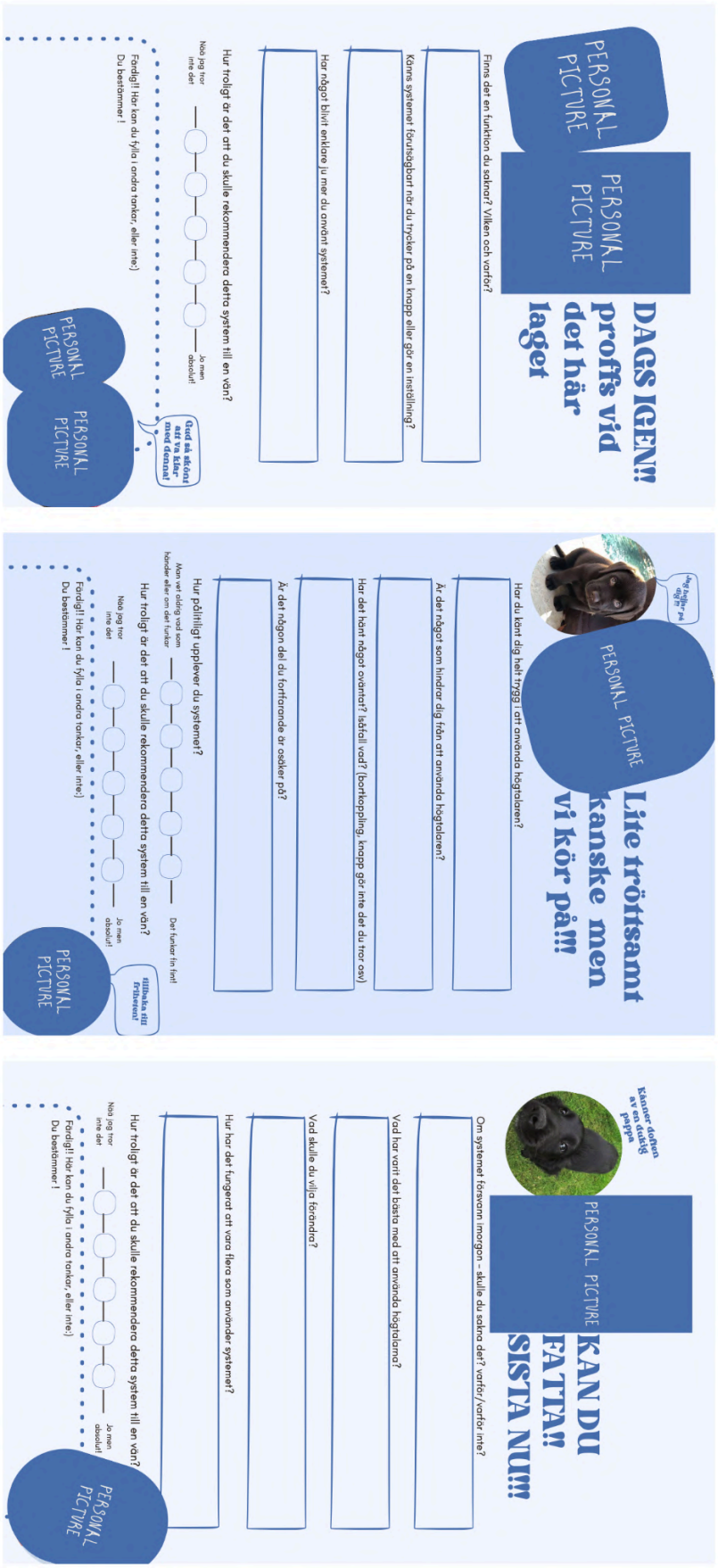



*Figure 2: The last three personalised letters for participant two*

At the office setting, the information was gathered through a thoughtsheet, see Figure 3, that was placed on a door in a passageway, to increase the likelihood that participants would interact with the observation material and leave feedback. Additionally, information at the office was also gathered through a letterbox. An installment with a letterbox, notepad, pen and accompanying questions were placed at a small table. Supportive questions that could be used to guide participant's thoughts when leaving feedback in the letterbox were prepared to lower the threshold for contribution. The questionsheet is presented in Figure 4.




*Figure 3: Thought sheet placed in the office*




*Figure 4: Supportive questions that could be used when leaving thoughts in the letterbox*

# Appendix B: Semi Structured Interview Guide

In the following chapter, the semi structured interview used during the interviews are presented. Since all participating interviewees were Swedish, the interview guide was made in Swedish. Questions considered especially important are marked in bold. If the interview guide deviated from the guide, the bold questions were always important to ask.

## Formella frågor

Är det okej om vi spelar in?
Är det okej att publicera dina svar i vår rapport?

## Introduktion (5 min)

Vad heter du?
Hur gammal är du?
Genom vad lyssnar du mest på musik när du är hemma? (högtalare, mobil, hörlurar)
    Varför brukar du lyssna på just detta sätt?
**I vilka situationer brukar du lyssna på musik? (ensam, vänner, ect)**
Hur skulle du beskriva din musiksmak?

## Teknikvana (10 min)

Hur tekniskt van skulle du beskriva dig själv?
Hur bekväm är du med att använda ny teknik?
**Hur snabb är du med att testa ny teknik?** (Är du några av de första, väntar du tills det är med etablerat på marknaden?)
    **Väljer du det sätt du redan vet av bekvämlighet?**
Hur brukar du lära dig ny teknik?
**Vad för anledningar finns det till att du inte skulle använda ny teknik?**
Vilka problem kan du stöta på när du testat ny teknik ? (kan vara ny teknik för dig)
    Hur löste du problemet?
Hur viktigt är det att ny teknik är lätt förstå direkt?
Finns det faktorer i ny teknik som hjälper dig att anamma det lättare?

## Högtalaren idag (10 min)

**Vad har du för högtalare idag?** (Märke, uppkoppling, ekosystem?)
Hur ofta använder du den?
Hur länge har du haft dem?
**Vad är anledningen till att du valde just den högtalaren?**
Stod du mellan flera olika när du skulle välja?
    Fanns det någon faktor som var övervägande vid valet?
Hur nöjd är du med högtalare idag?
    Varför är du nöjd/inte nöjd?
**Vad är ett återkommande problem för dig med användandet?**

## Högtalares miljö (5 min)

Är högtalaren en teknikprodukt eller en inredningsdetalj för dig?
**Brukar din högtalare röra sig utanför ditt hem? Isåfall vart?**
**Vad är viktigt för dig om man ska ta med högtalare ut?**
Finns det något som hindrar dig från att ta med högtalaren ut?

## Medvetenhet om val (10min)

**När du spelar musik, hur sätter du på din musik?** (genom inställning, airplay osv)
Varför använder du sättet?
Om flera personer ska spela musik på samma högtalare, hur brukar det gå till?
**Vilka olika sätt känner du till för att spela musik på en högtalare?**
Hur skiljer de sig åt? Vad är fördelen/nackdelen med en viss metod?

## Avslut

Har intervju väckt några frågor eller åsikter? vilka isåfall?
Något mer att tillägga?

# Appendix C: Storyboard

Based on the personas, a storyboard was made to identify events and behaviours surrounding the speaker that may occur during a day. The storyboard can be found in Figure 1.

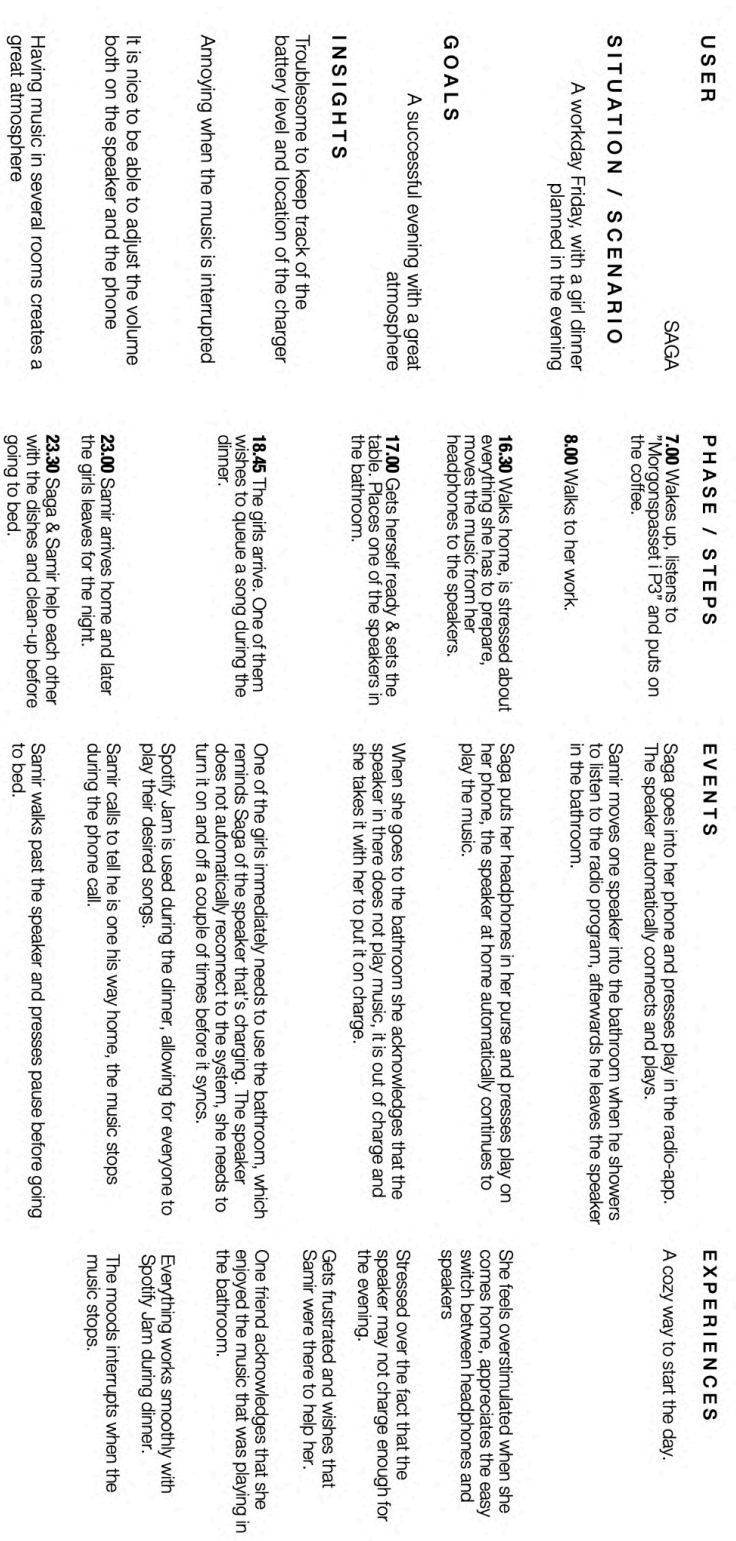



*Figure 1: Storyboard of Persona Saga*

# Appendix D: Workshop

A workshop was conducted with eight participants to explore what users precise as intuitive design in an application. The participants were divided into two groups.

During the first part of the workshop, the participants were asked to design application pages based on a series of instructions. Both groups were provided with a large grey paper sheet, representing a phone screen, along with smaller elements that could be placed on the screen as desired. The first task was to design a homescreen. The resulting designs produced by each group are presented in Figure 1.

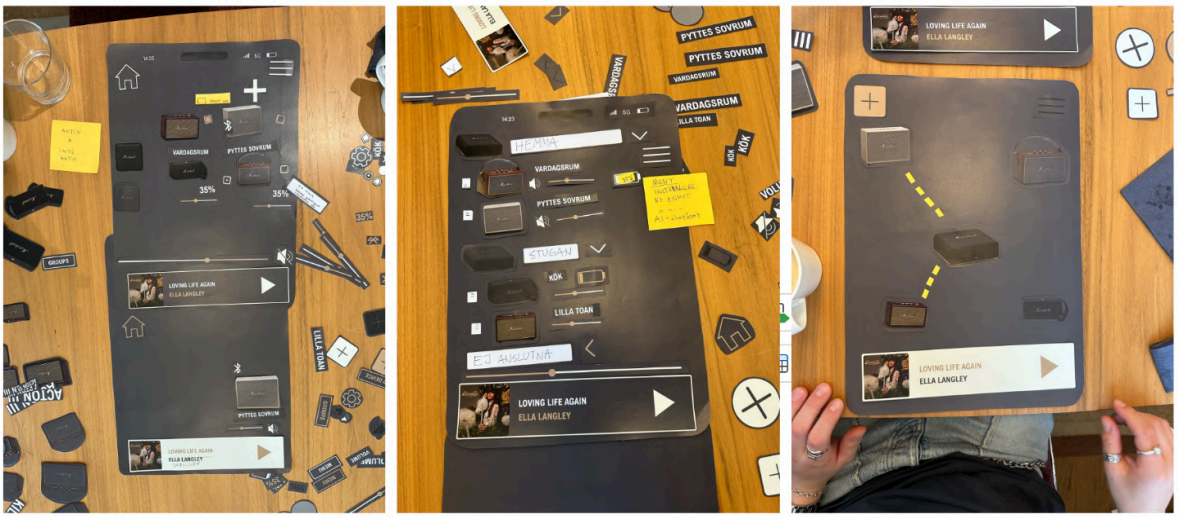



*Figure 1: Layout of the homescreen. Screen to the left is group 1, middle screen and right screen are group 2.*

After completing the homescreen design, the participants were presented with different usage scenarios and asked how these situations should be visually represented on the home screen. Scenario 1, illustrated in Figure 2, represents a situation in which Acton loses its connection to Heddon. Scenario 2, shown in Figure 3, illustrates how the interface should indicate that a portable speaker is out of battery, in this case, the affected device is Kilburn. The final scenario, scenario 3, presents the homescreen when Heddon has lost its connection, shown in Figure 4.

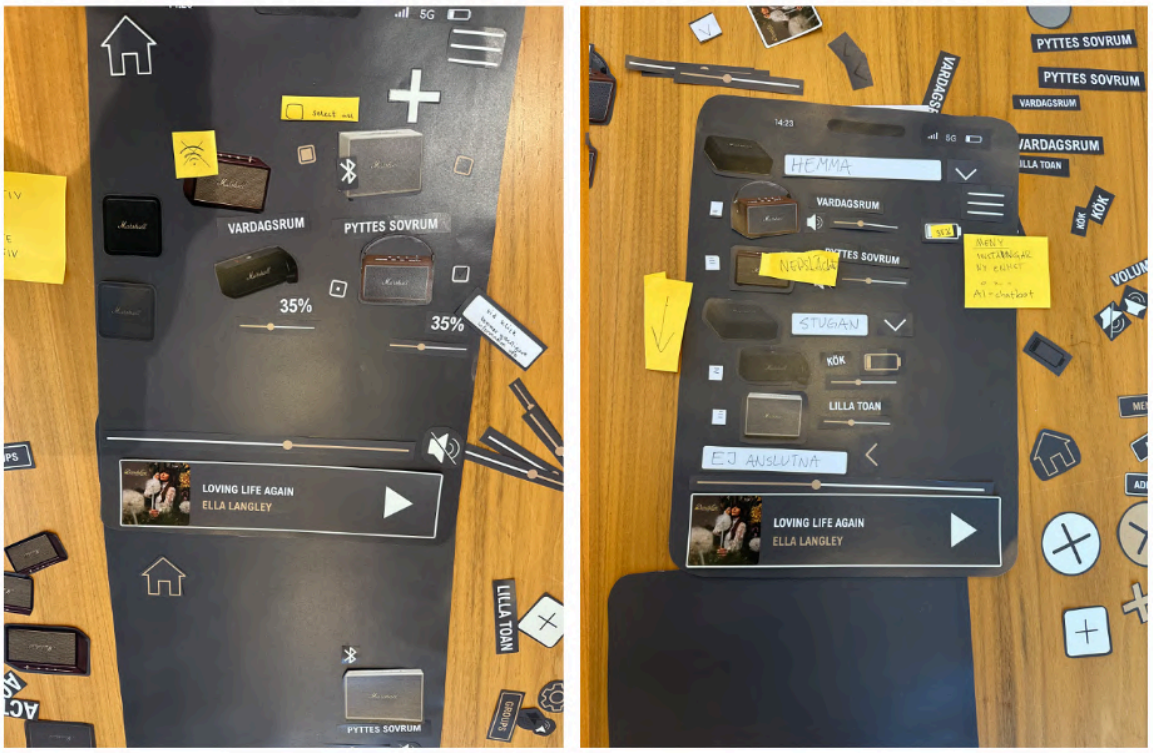



*Figure 2: Layout of homescreen if Acton loses its connection. Screen to the left is group 1 and screen to the right is group 2.*

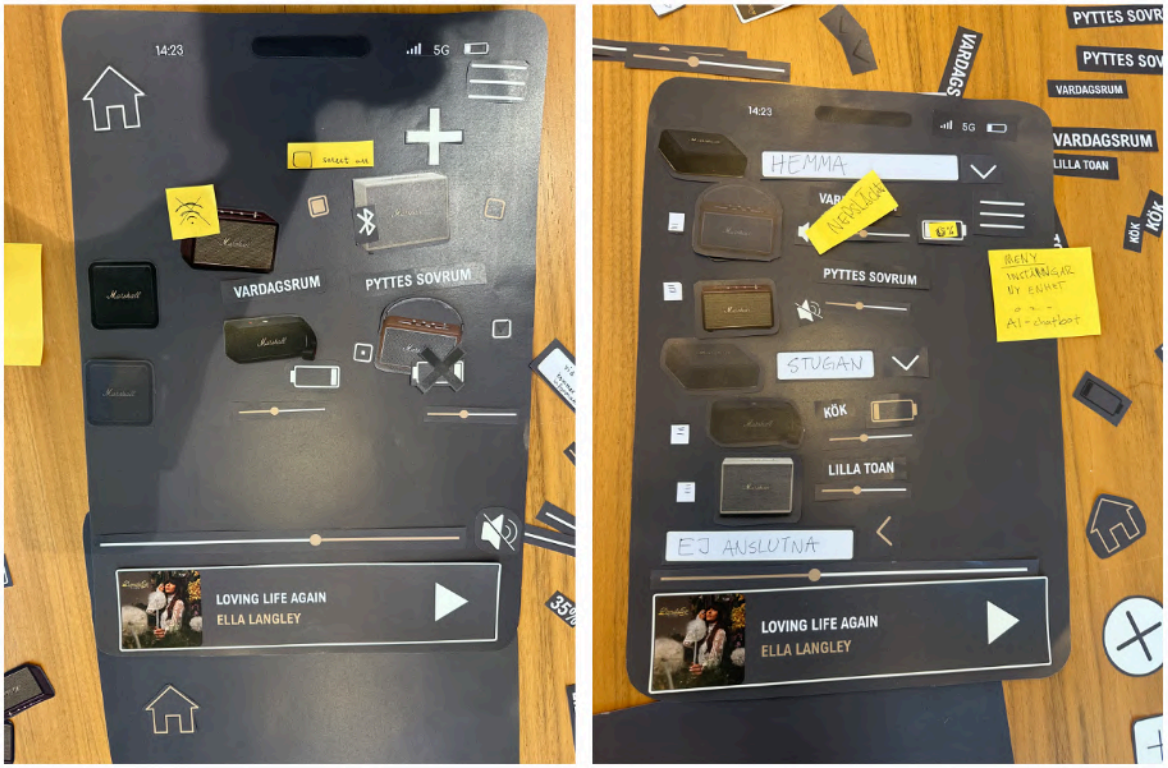



*Figure 3: Homescreen if Kilburn is out of battery. Screen to the left is group 1 and screen to the right is group 2.*

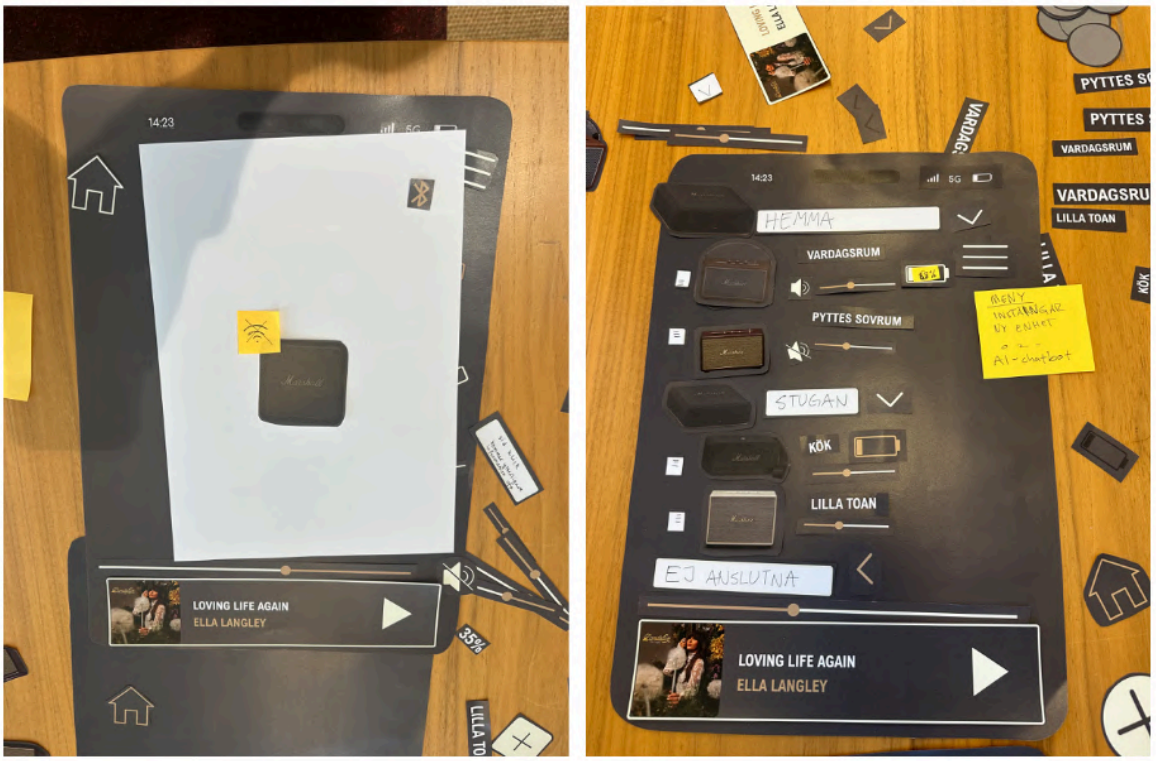



*Figure 4: Homescreen if Heddon lost its connection. Screen to the left is group 1 and screen to the right is group 2.*

Following the scenarios, the participants were asked to design an individual page for a single device.

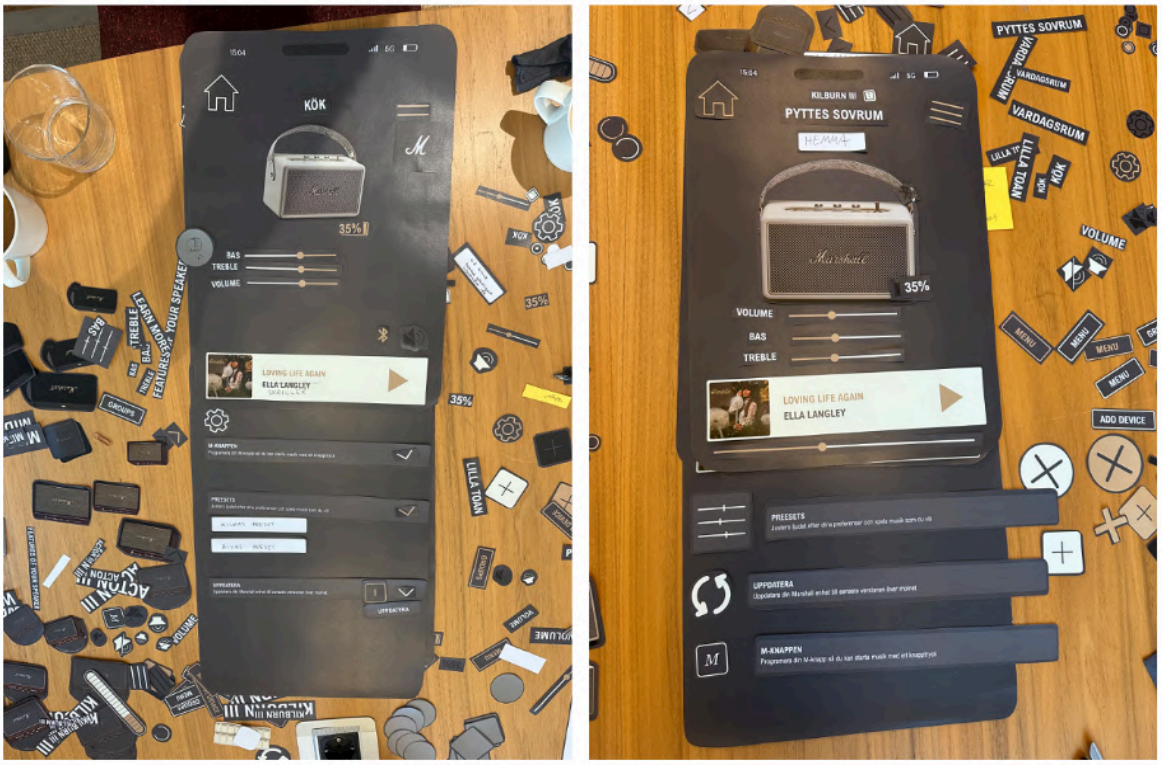



*Figure 5: Individual device page. Screen to the left is group 1and screen to the right is group 2.*

# Appendix E: First Draft of Application Layout

The first drafts of the application are presented in the following chapter. The layout of the application was derived from the results of the workshop. Figure 1 visualises the homescreen with all speakers installed as well as when Heddon is one of the installed units.




*Figure 1: Homescreen without Heddon system to the left and with Heddon on the right*

Figure 2 presents three potential cases that can occur during use with the system. Case one presents the layout of the homescreen if a device loses its connection from the system. Case two visualises what it looks like if one portable unit is out of charge and therefore are not a part of the system anymore. Lastly, case three presents what the homescreen would look like if Heddon lost its connection.




*Figure 2: Visualisation of layout of the three scenarios, from the right: case one, case two and case three*

Then a draft of a page with an individual device was proposed. Two different layouts showing the device's status to Heddon were tried out. Further, visualisation of the individual page devices if scrolled down is also presented.

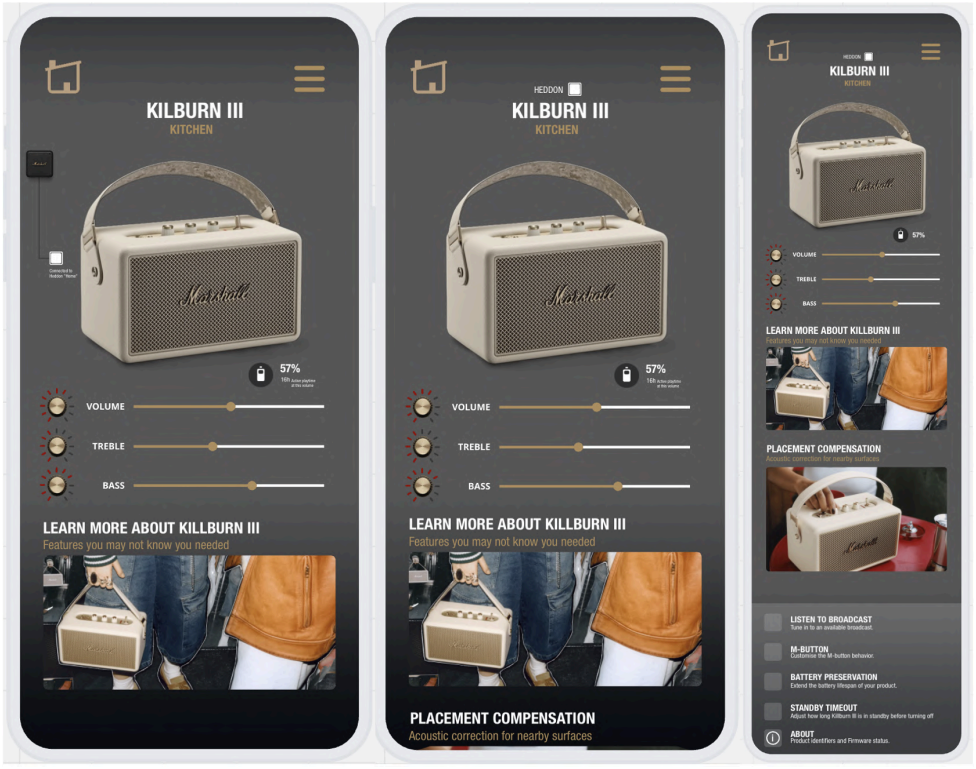



*Figure 3: Two examples of layouts for individual device page with connection to Heddon system*

Lastly, a menu page was configured illustrated. Some new categories were added while maintaining Marshall's identity. A first proposed layout is shown in Figure 4.

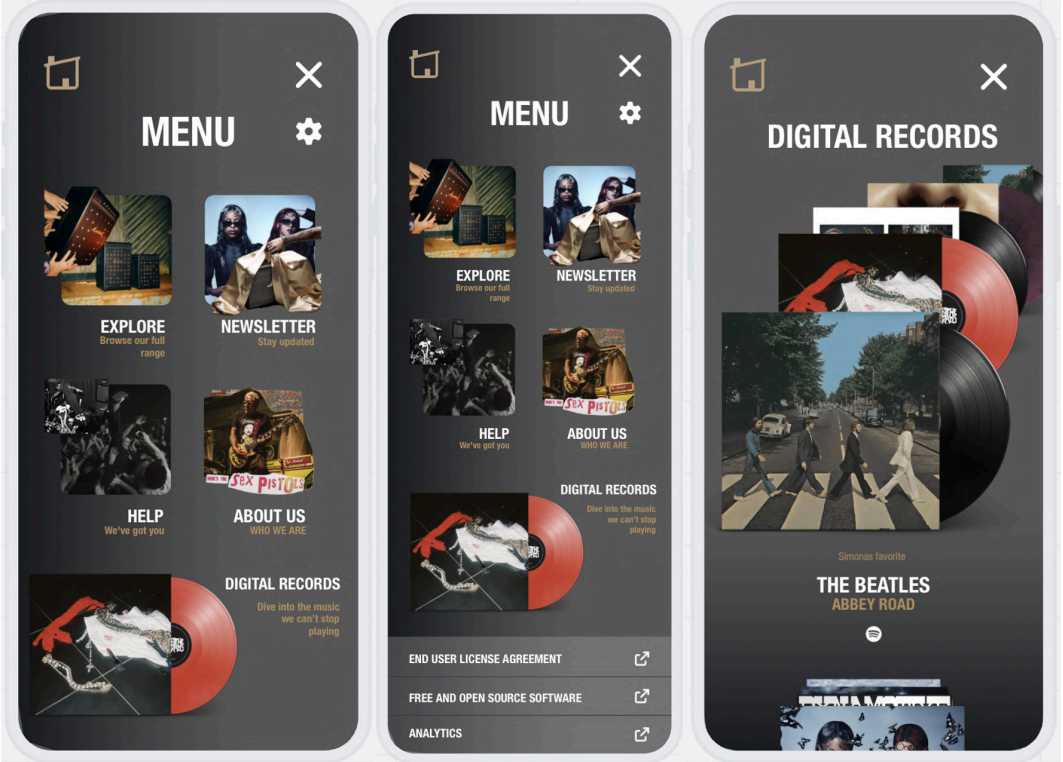



*Figure 4: Overview of Menu with new categories*

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